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PV shading models with pvlib

POWERED BY

Accelerating photovoltaic system simulations via statistical data aggregation

Adrián Blanco Aguiar

Simulation process

Hourly operating conditions

Operating conditions

Temperature



Sun position



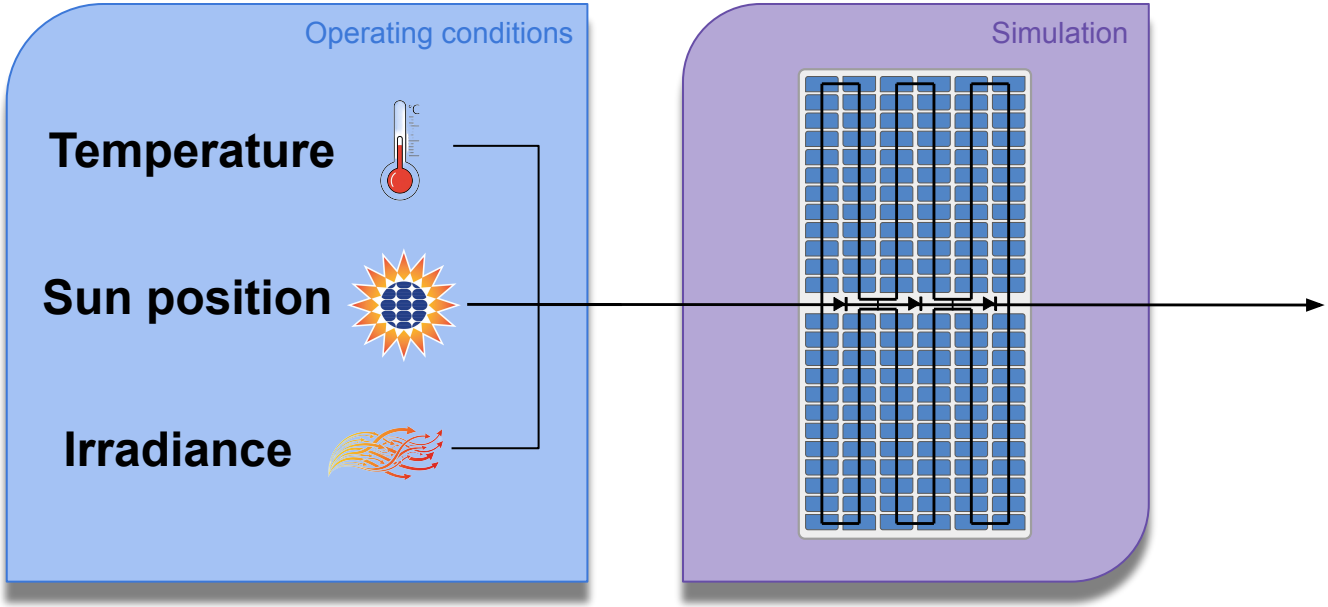
Irradiance



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Simulation process

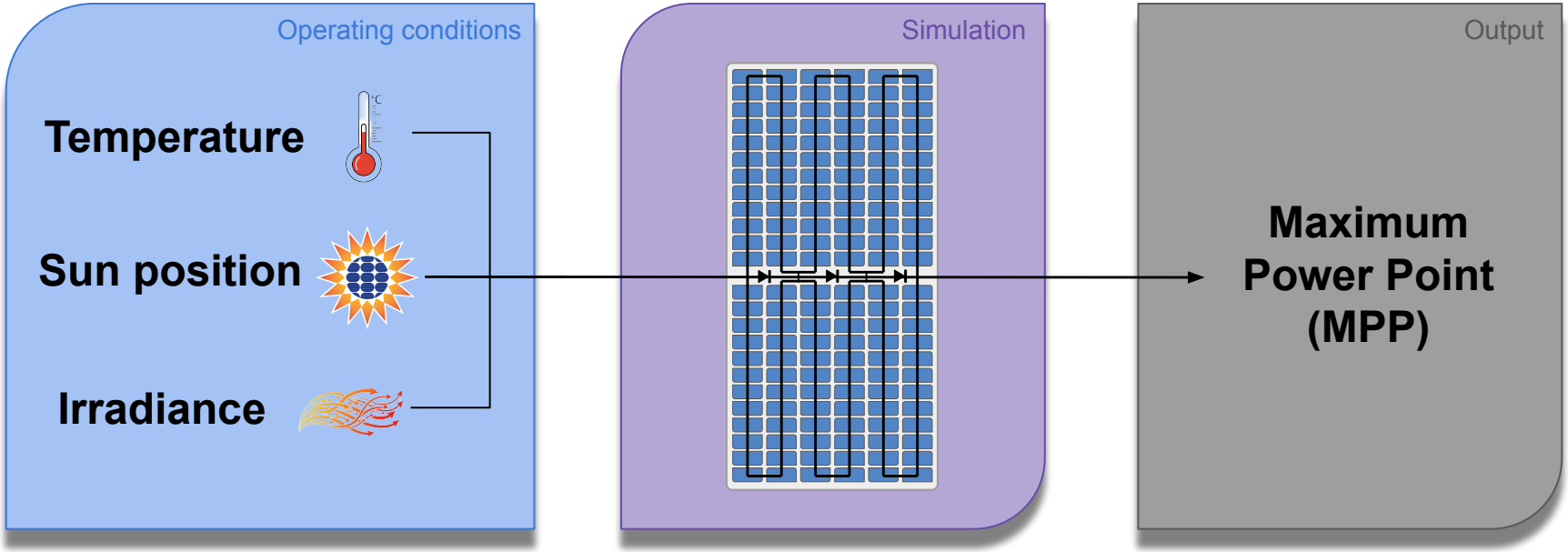
Hourly operating conditions



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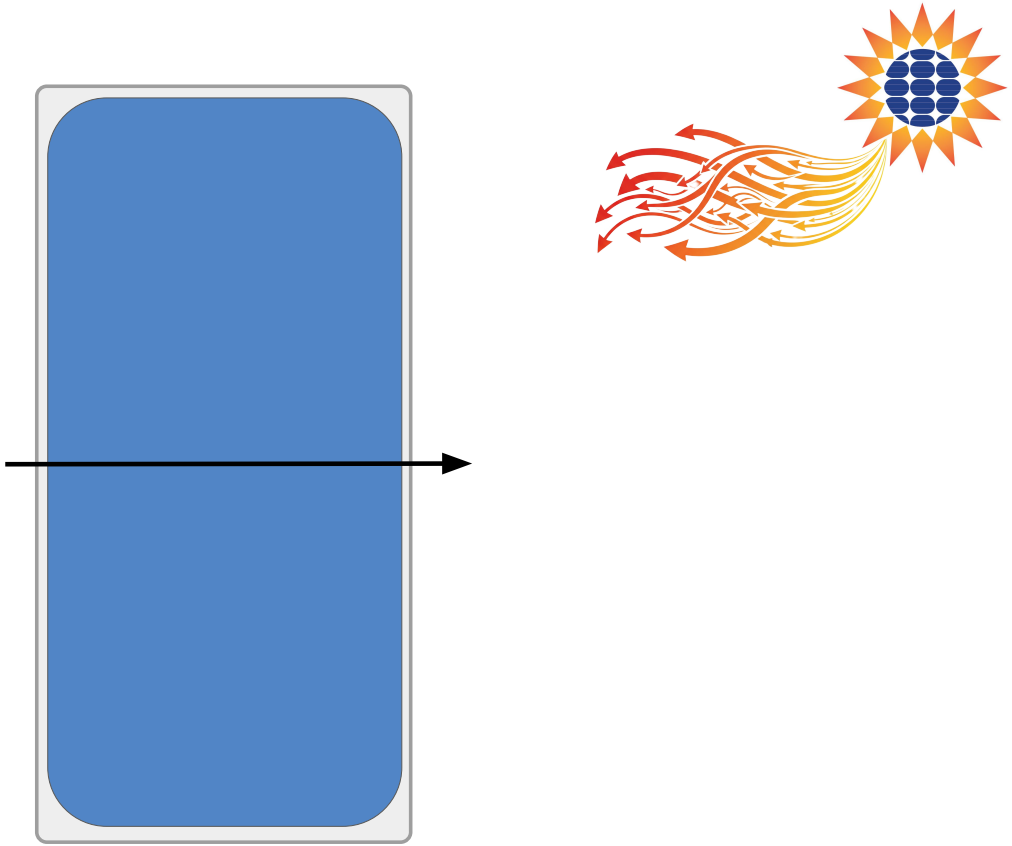
Simulation process

Hourly operating conditions



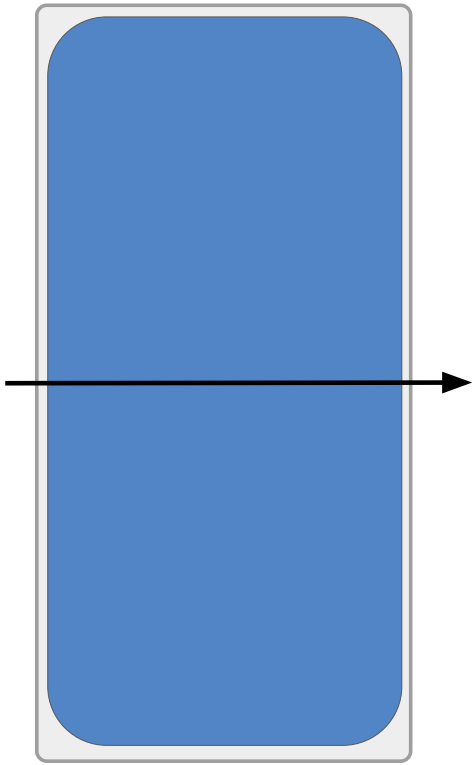
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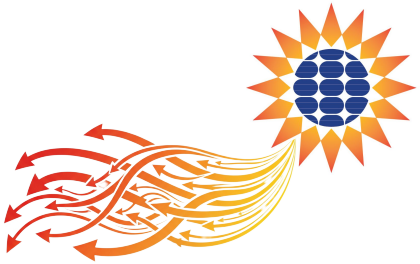
Module-level without considering shading

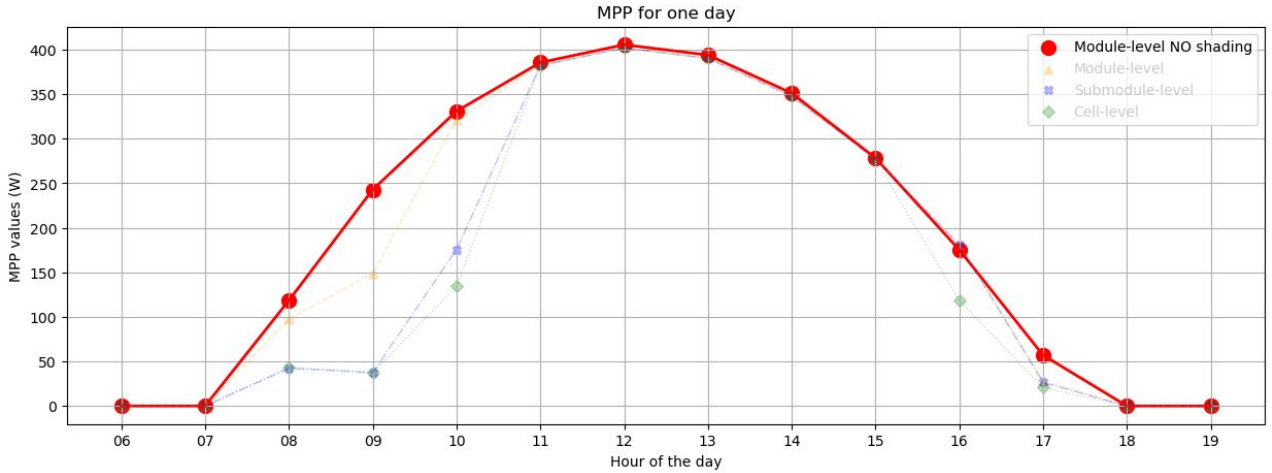


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Module-level without considering shading

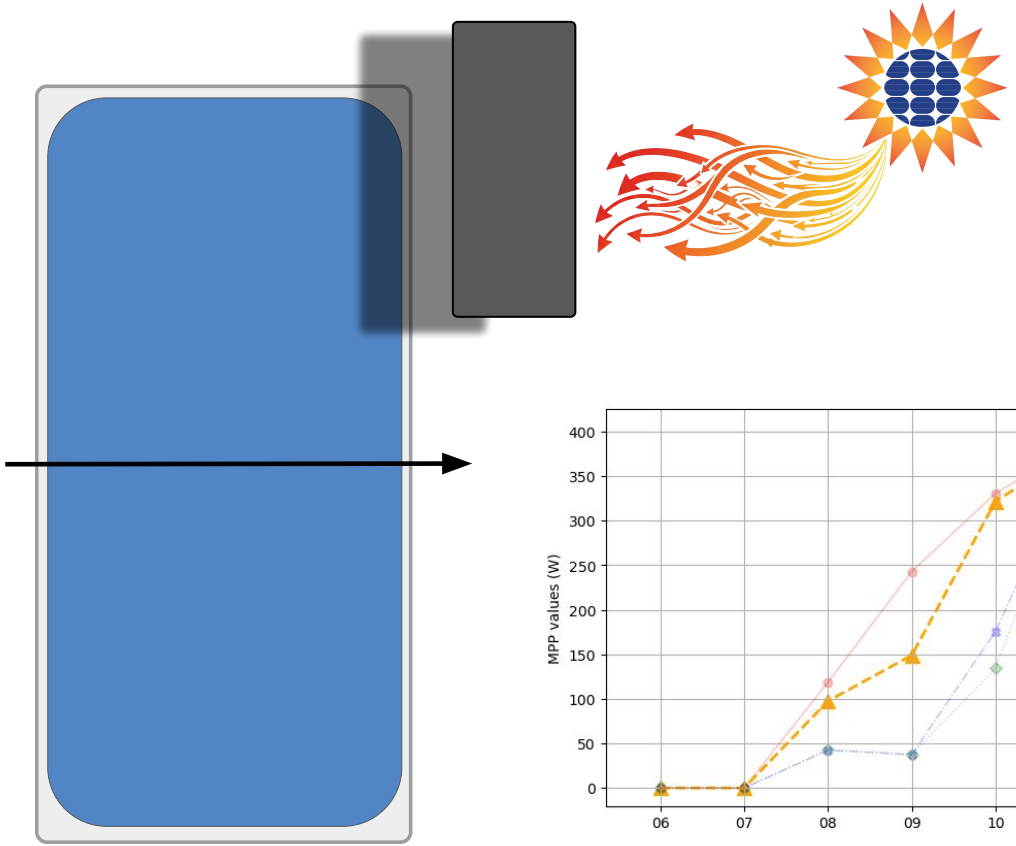


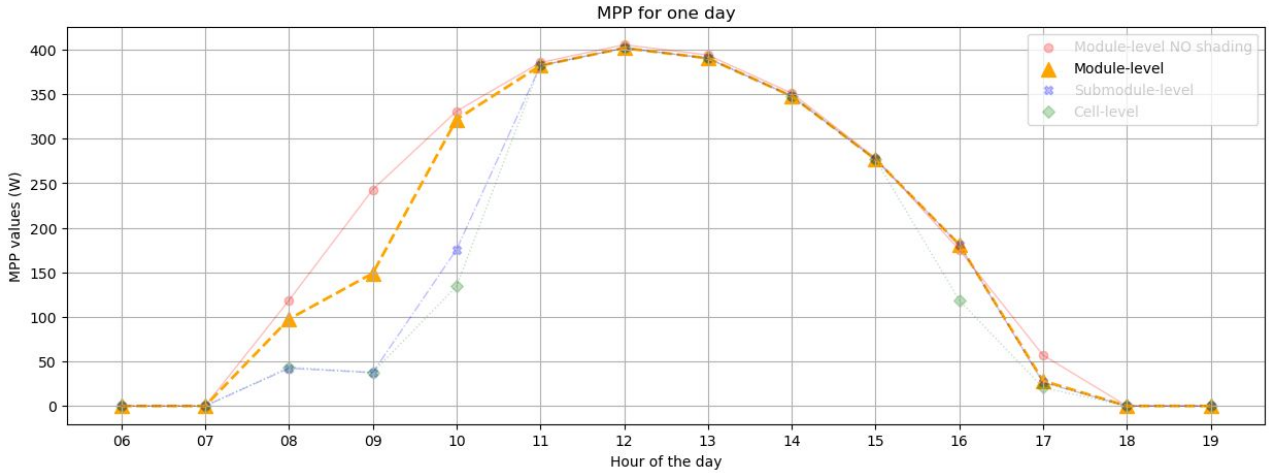




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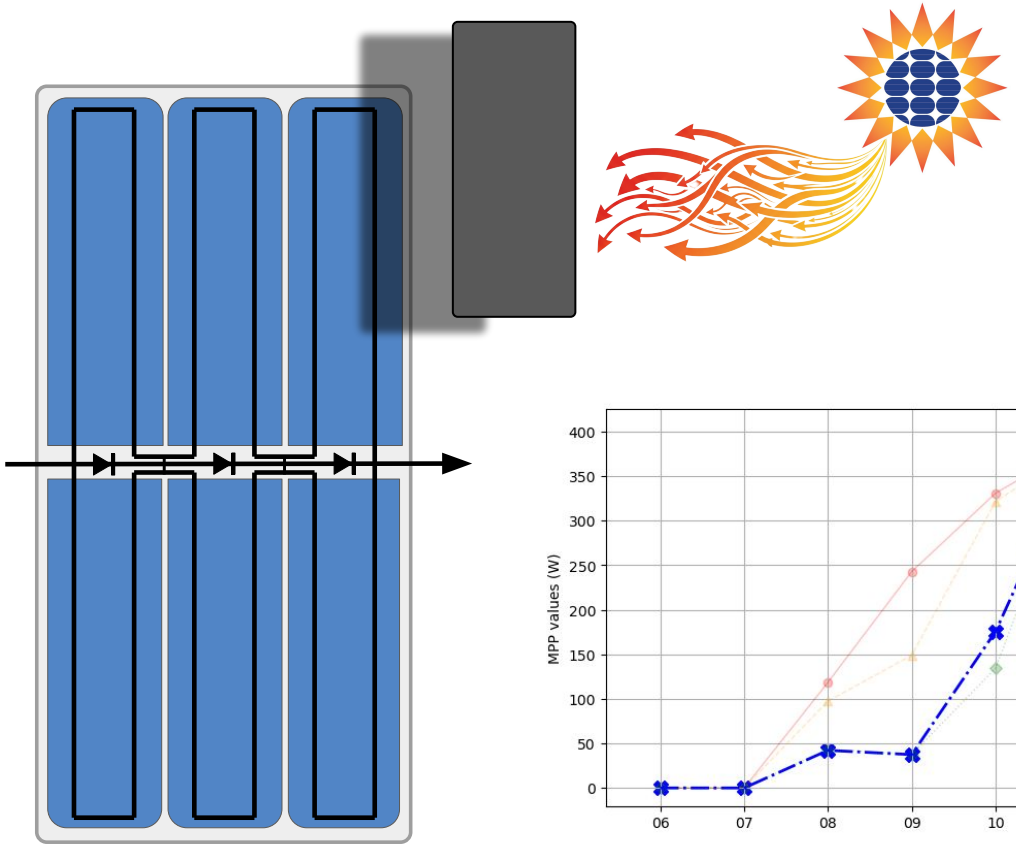
Module-level

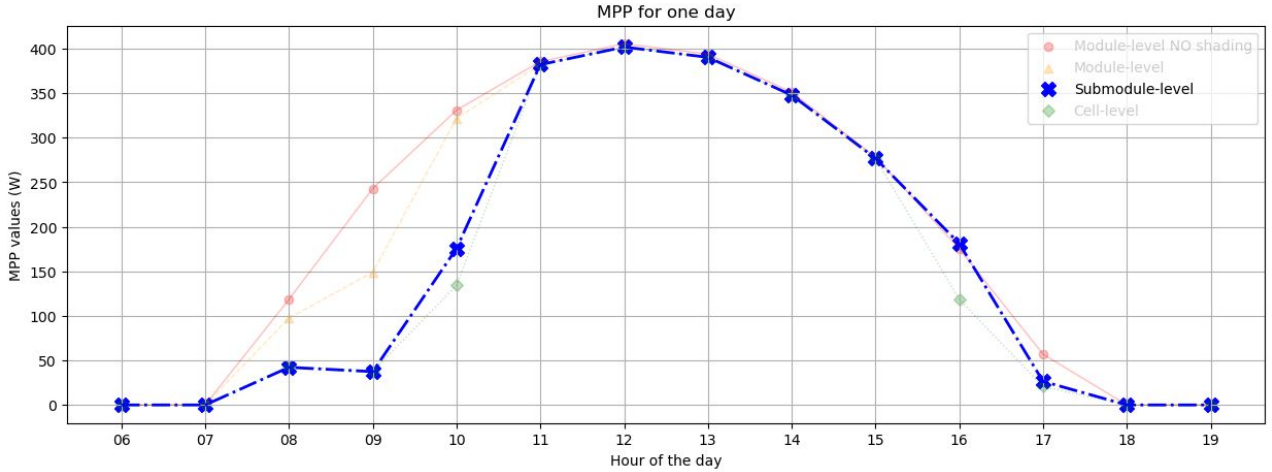




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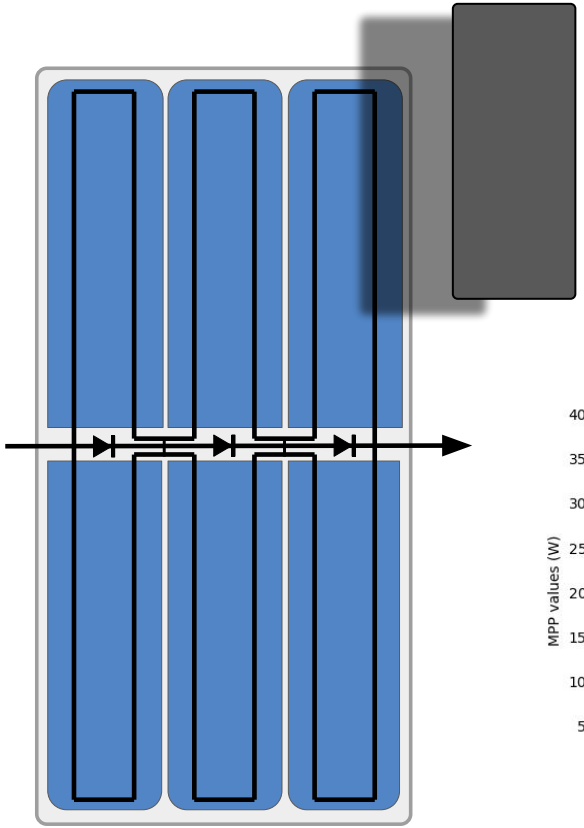
Submodule-level

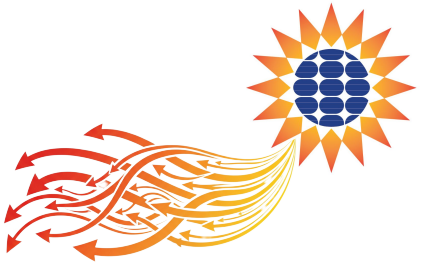




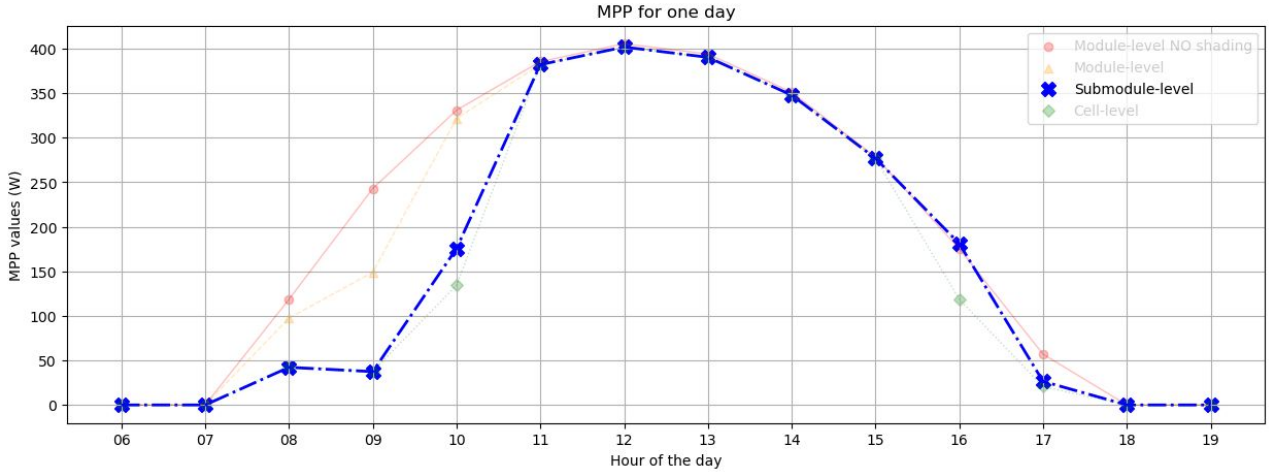
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Submodule-level





This is the one we use as the accurate reference method in our work!



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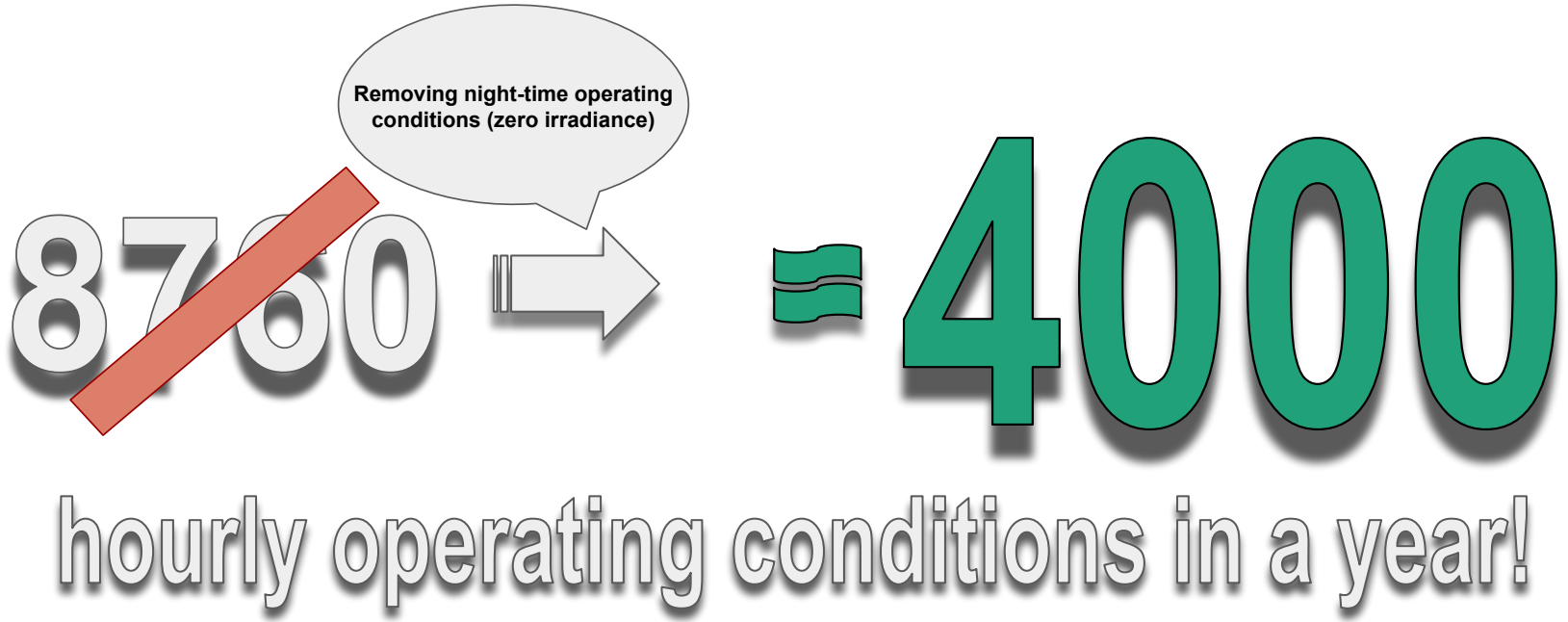
Problem

8760

hourly operating conditions in a year!

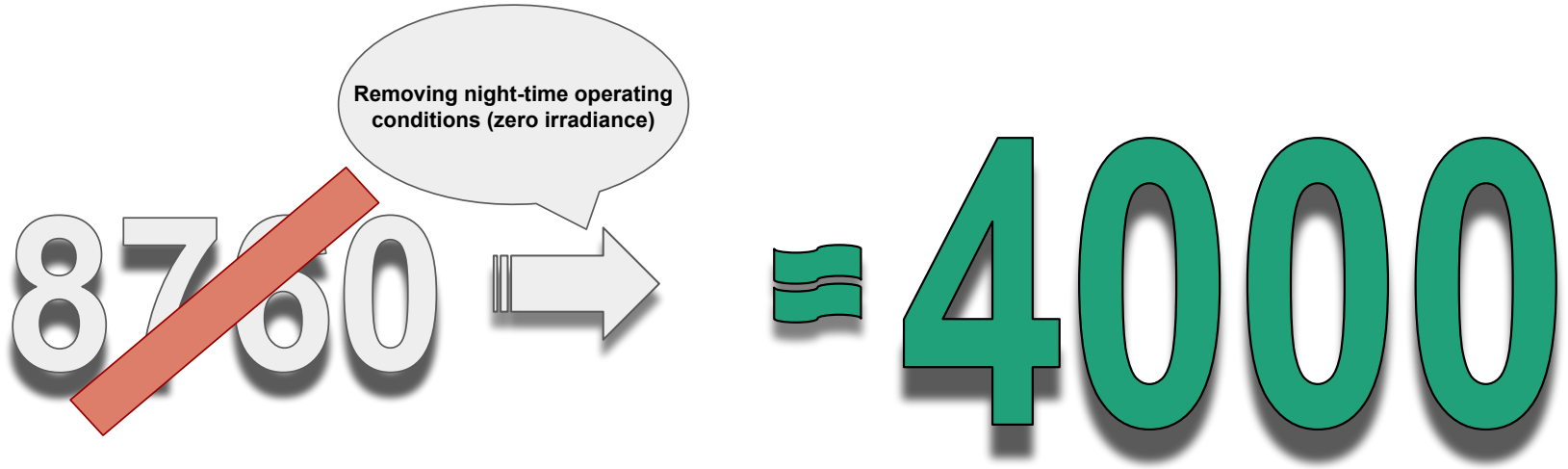
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Problem



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Problem



hourly operating conditions in a year!

high computational cost

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4000

How do we reduce this number?

Similar operating conditions produce similar results

2 November, 11:00



Irradiance



Sun elevation

Sun azimuth



Temperature

3 November, 11:00



Irradiance



Sun elevation

Sun azimuth



Temperature

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Similar operating conditions produce similar results

2 November, 11:00

Irradiance ⟹ 922.49 W/m²

Sun elevation

Sun azimuth

Temperature

3 November, 11:00

Irradiance ⟹ 919.51 W/m²

Sun elevation

Sun azimuth

Temperature

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Similar operating conditions produce similar results

2 November, 11:00

Irradiance	⟹	922.49 W/m²
Sun elevation	⟹	31.14°
Sun azimuth		
Temperature		

3 November, 11:00

Irradiance	⟹	919.51 W/m²
Sun elevation	⟹	30.85°
Sun azimuth		
Temperature		

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Similar operating conditions produce similar results

2 November, 11:00

Irradiance	⟹	922.49 W/m²
Sun elevation	⟹	31.14°
Sun azimuth	⟹	-24.87°
Temperature		

3 November, 11:00

Irradiance	⟹	919.51 W/m²
Sun elevation	⟹	30.85°
Sun azimuth	⟹	-24.75°
Temperature		

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Similar operating conditions produce similar results

2 November, 11:00

Irradiance	⟹	922.49 W/m²
Sun elevation	⟹	31.14º
Sun azimuth	⟹	-24.87º
Temperature	⟹	43.19ºC

3 November, 11:00

Irradiance	⟹	919.51 W/m²
Sun elevation	⟹	30.85º
Sun azimuth	⟹	-24.75º
Temperature	⟹	43.39ºC

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Similar operating conditions produce similar results

2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

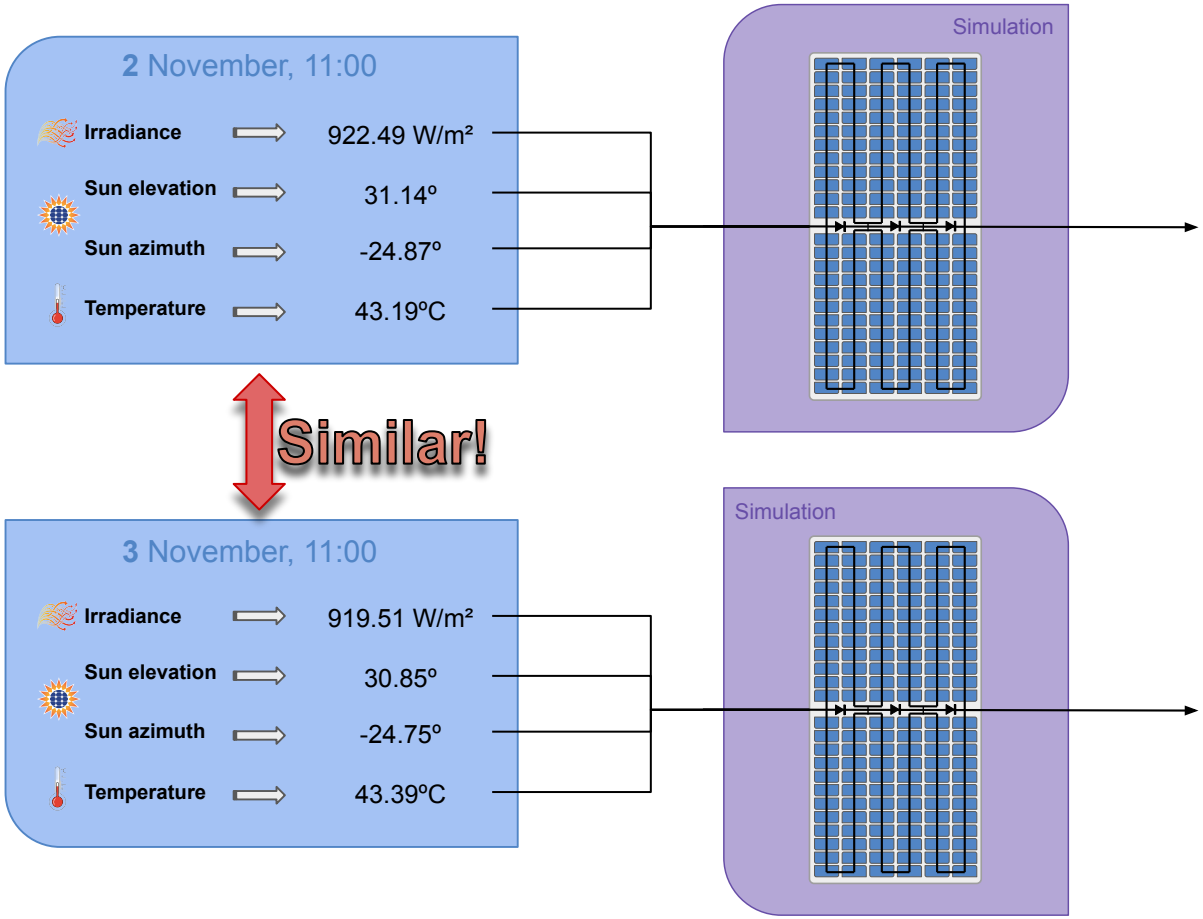
Similar!

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

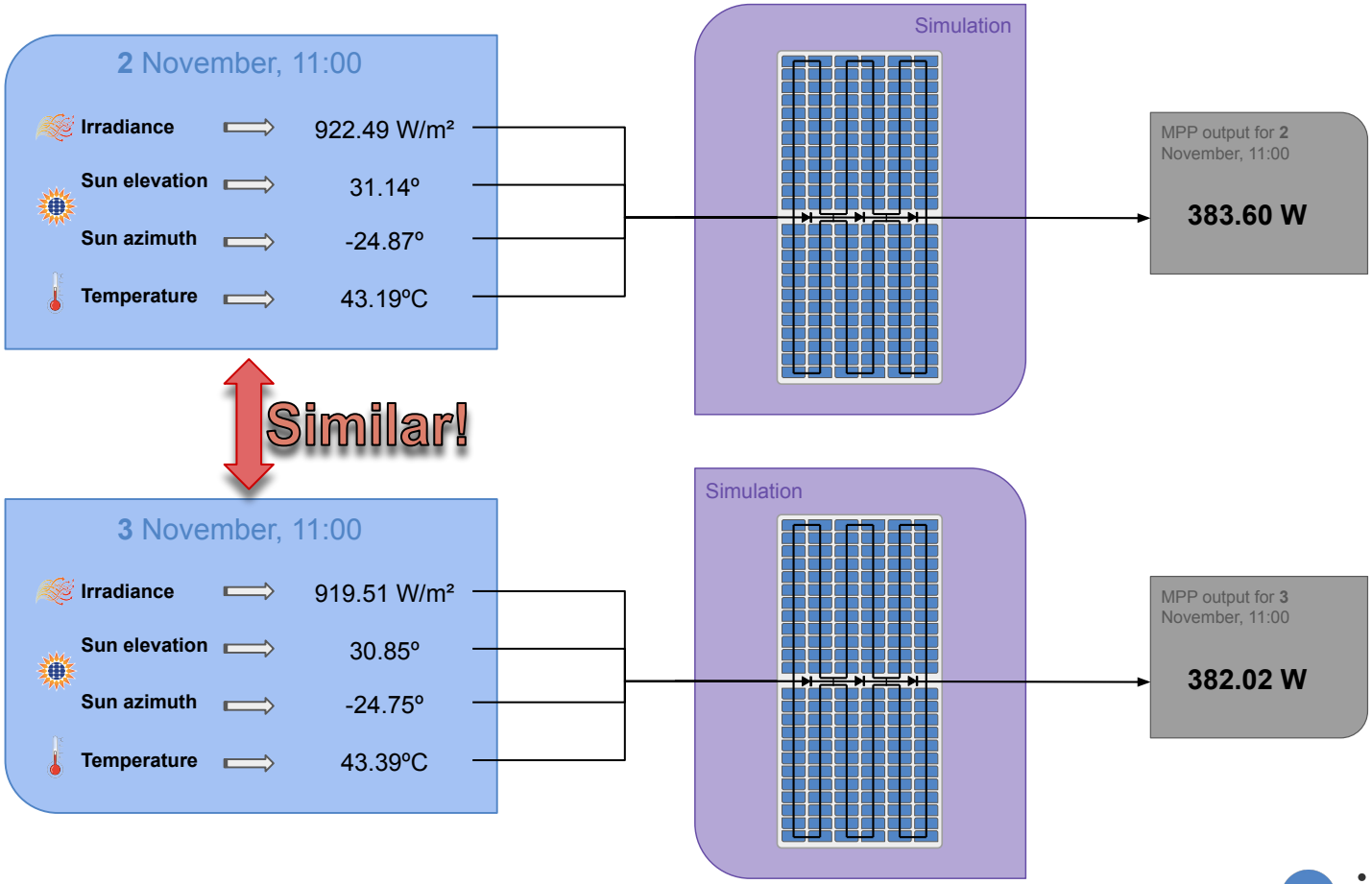
ieco.io

Similar operating conditions produce similar results



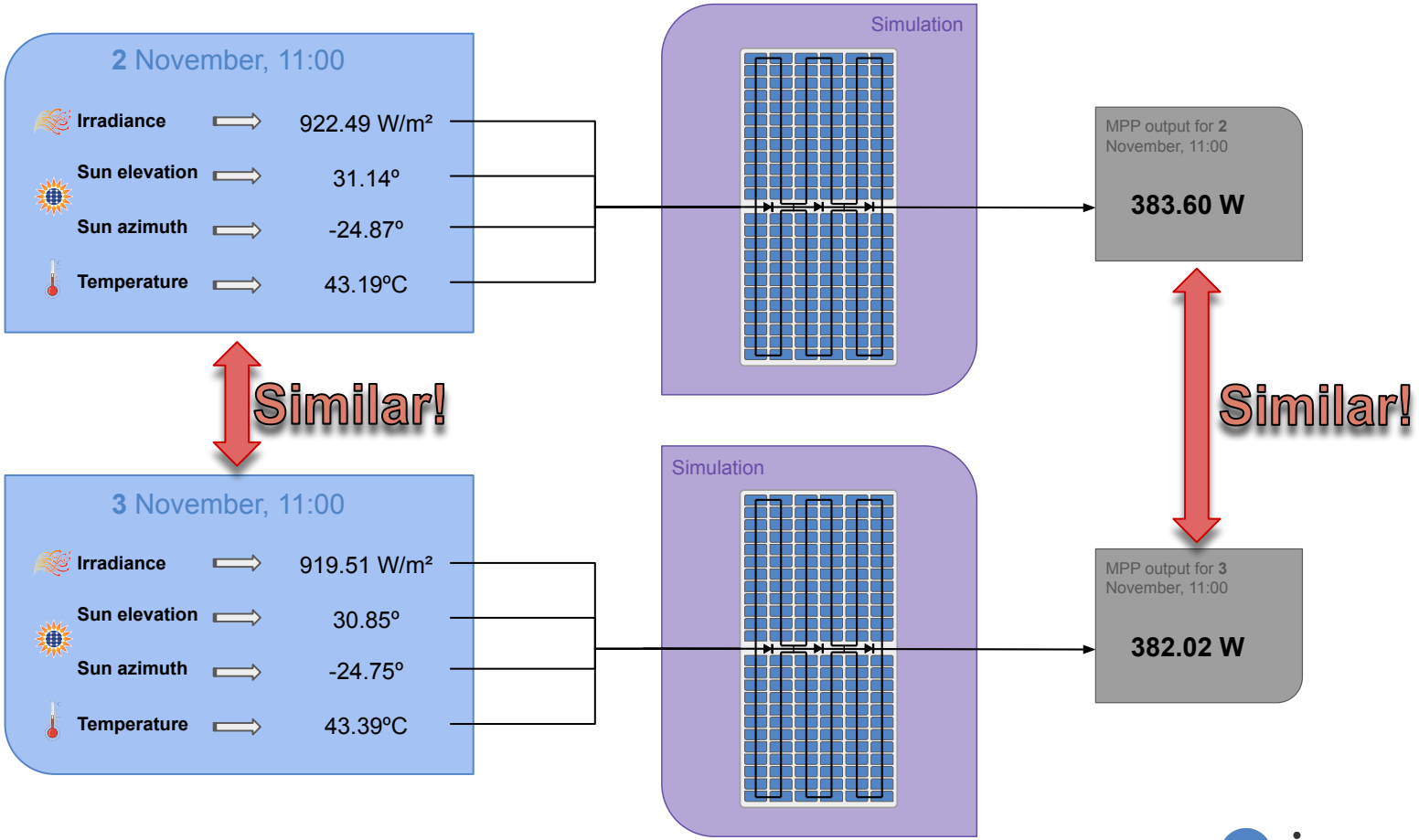
ieco.io

Similar operating conditions produce similar results



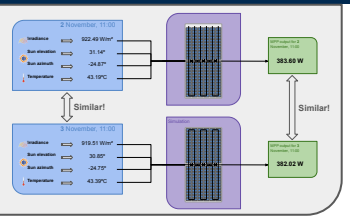
ieco.io

Similar operating conditions produce similar results



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Aggregation of similar operating conditions



2 November, 11:00

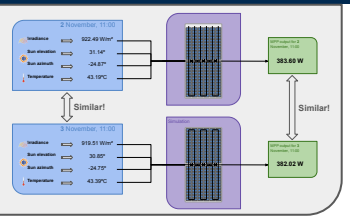
Irradiance	⟹	922.49 W/m²
Sun elevation	⟹	31.14°
Sun azimuth	⟹	-24.87°
Temperature	⟹	43.19°C

3 November, 11:00

Irradiance	⟹	919.51 W/m²
Sun elevation	⟹	30.85°
Sun azimuth	⟹	-24.75°
Temperature	⟹	43.39°C

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Aggregation of similar operating conditions



2 November, 11:00

Irradiance	⇨	922.49 W/m²
Sun elevation	⇨	31.14°
Sun azimuth	⇨	-24.87°
Temperature	⇨	43.19°C

3 November, 11:00

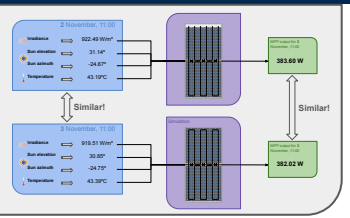
Irradiance	⇨	919.51 W/m²
Sun elevation	⇨	30.85°
Sun azimuth	⇨	-24.75°
Temperature	⇨	43.39°C

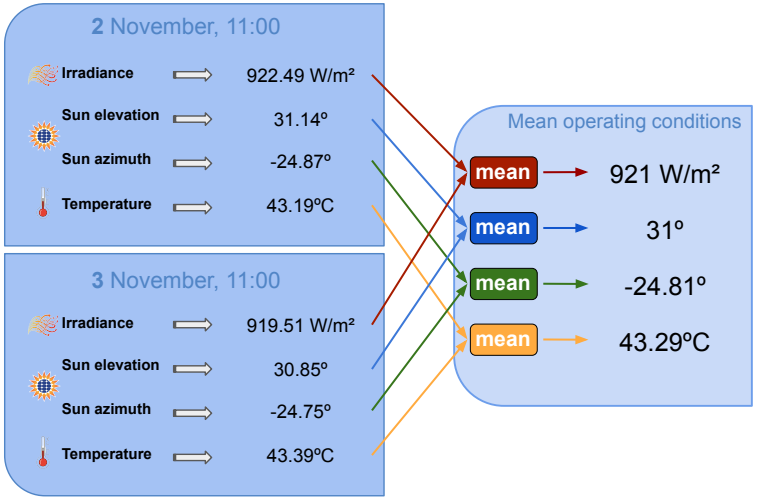
Mean operating conditions

mean → 921 W/m²

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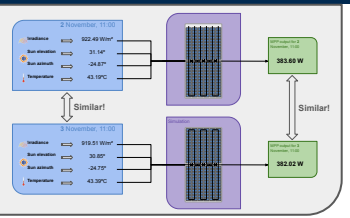
Aggregation of similar operating conditions

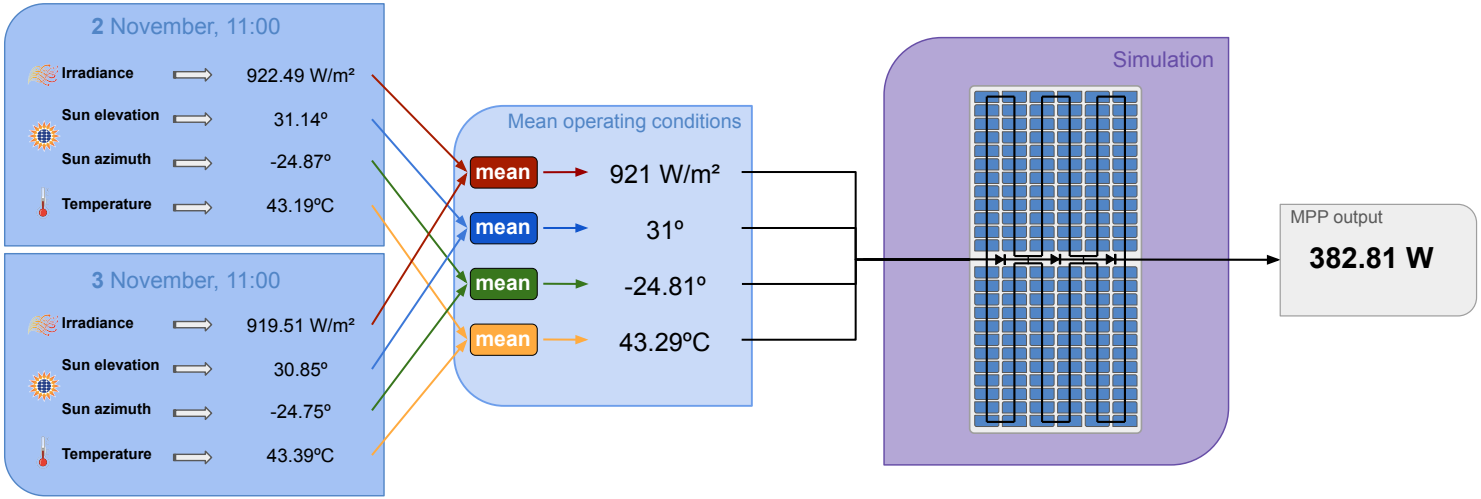




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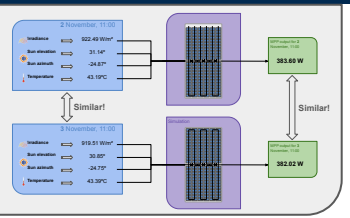
Aggregation of similar operating conditions





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Aggregation of similar operating conditions



2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

Mean operating conditions

mean	921 W/m²
mean	31°
mean	-24.81°
mean	43.29°C

Simulation

MPP output

382.81 W

MPP output for **2** November, 11:00

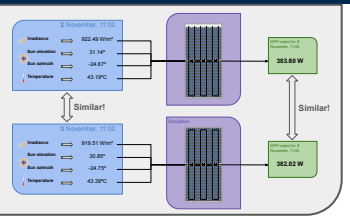
382.81 W

MPP output for **3** November, 11:00

382.81 W

ieco.io

Aggregation of similar operating conditions



2 November, 11:00

- Irradiance → 922.49 W/m²
- Sun elevation → 31.14°
- Sun azimuth → -24.87°
- Temperature → 43.19°C

3 November, 11:00

- Irradiance → 919.51 W/m²
- Sun elevation → 30.85°
- Sun azimuth → -24.75°
- Temperature → 43.39°C

Mean operating conditions

- mean → 921 W/m²
- mean → 31°
- mean → -24.81°
- mean → 43.29°C

Simulation

MPP output

382.81 W

Real value: 383.60 W

MPP output for **2** November, 11:00

382.81 W

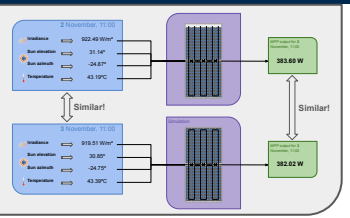
MPP output for **3** November, 11:00

382.81 W

Real value: 382.02 W

ieco.io

Aggregation of similar operating conditions



Just one simulation!

2 November, 11:00

Irradiance	922.49 W/m²
Sun elevation	31.14°
Sun azimuth	-24.87°
Temperature	43.19°C

3 November, 11:00

Irradiance	919.51 W/m²
Sun elevation	30.85°
Sun azimuth	-24.75°
Temperature	43.39°C

Mean operating conditions

mean	921 W/m²
mean	31°
mean	-24.81°
mean	43.29°C

Simulation

MPP output

382.81 W

Real value:
383.60 W

MPP output for **2** November, 11:00

382.81 W

MPP output for **3** November, 11:00

382.81 W

Real value:
382.02 W

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How do we automatically aggregate similar operating conditions?

Aggregation of similar operating conditions

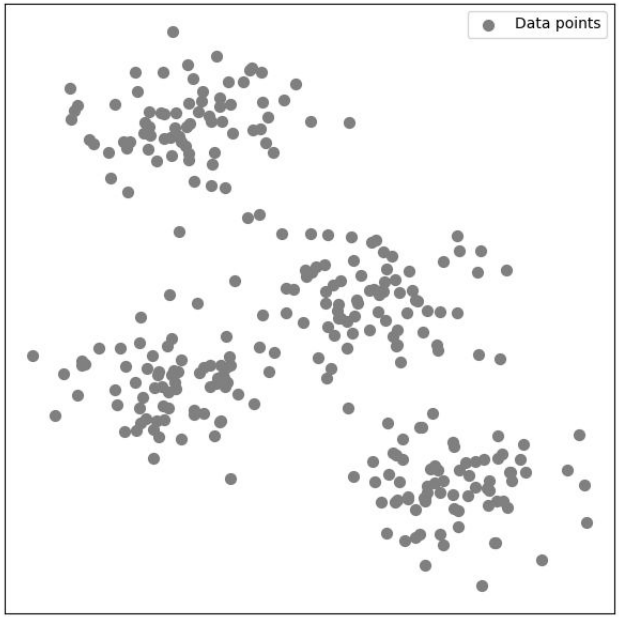
How do we automatically aggregate similar operating conditions?

k-means

Unsupervised machine learning algorithm for data segmentation (clustering)

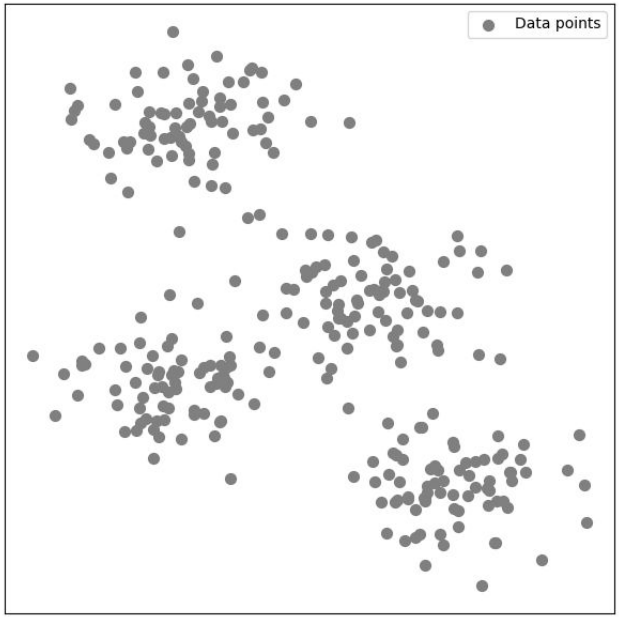
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K-means algorithm

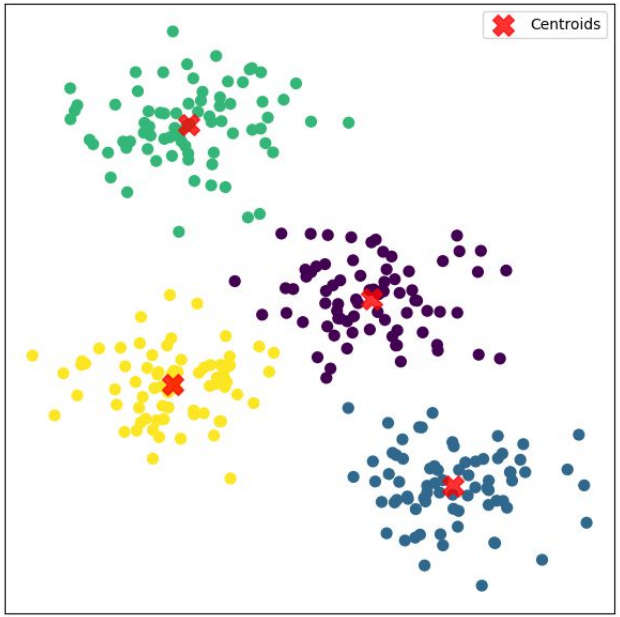


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K-means algorithm

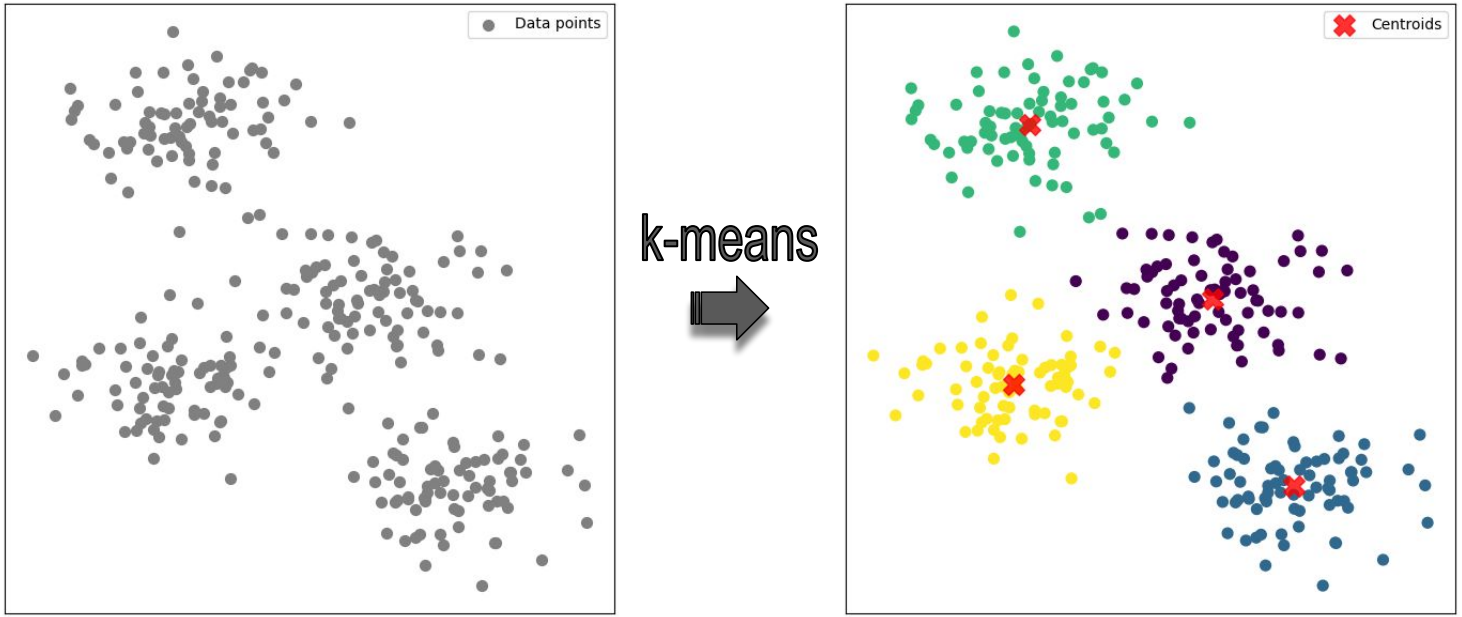


k-means



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K-means algorithm



Centroids (✖) as new data points

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Initial data aggregation approaches

- StraightForward Aggregation (SFA)

Initial data aggregation approaches

- StraightForward Aggregation (SFA)
- Hierarchical Hourly Aggregation (HHA)

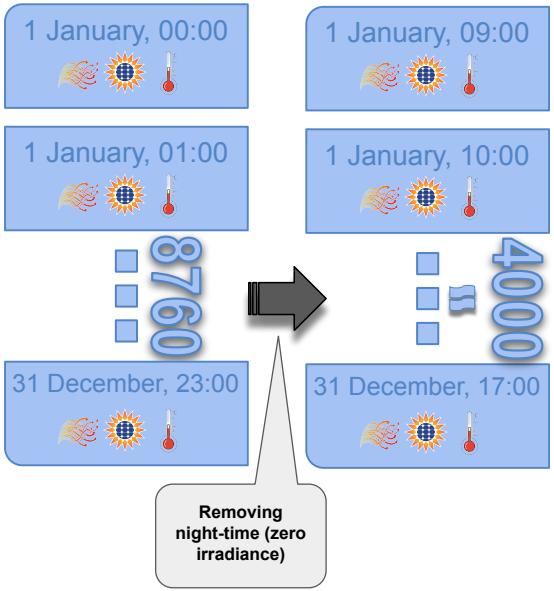
ieco.io

Initial data aggregation approaches

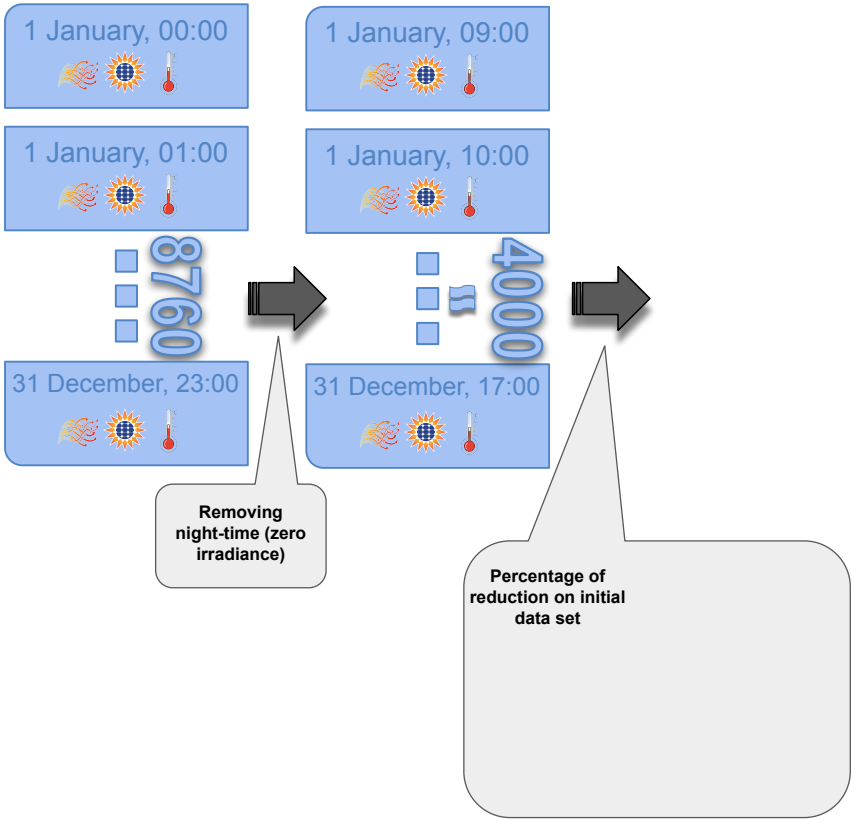
- StraightForward Aggregation (SFA)
- Hierarchical Hourly Aggregation (HHA)

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Straight-Forward Aggregation (SFA)

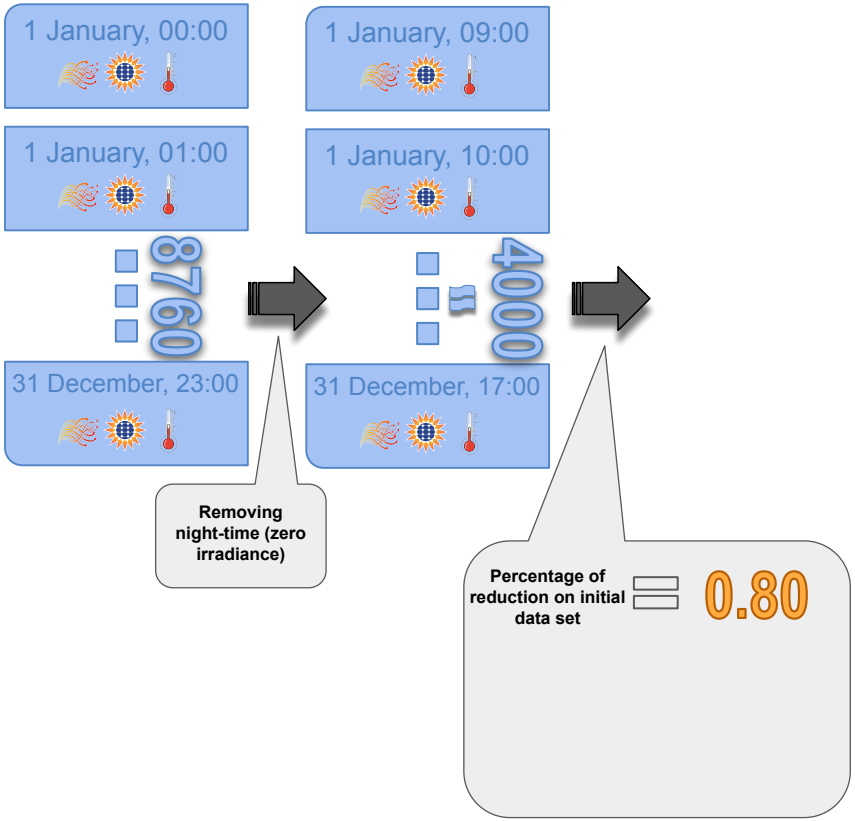


Straight-Forward Aggregation (SFA)



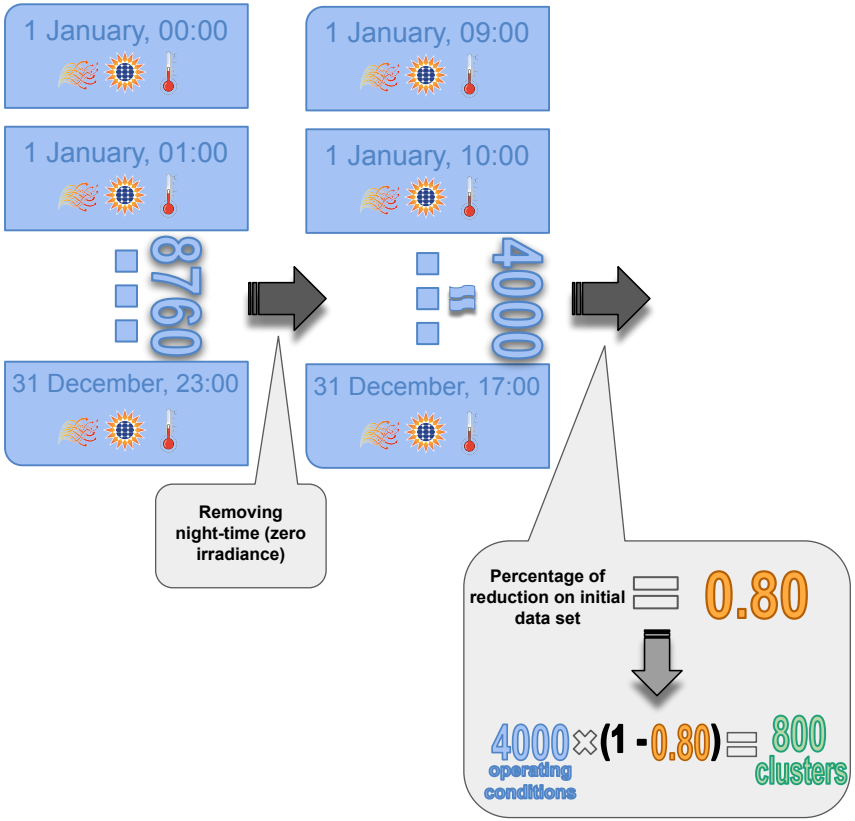
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Straight-Forward Aggregation (SFA)



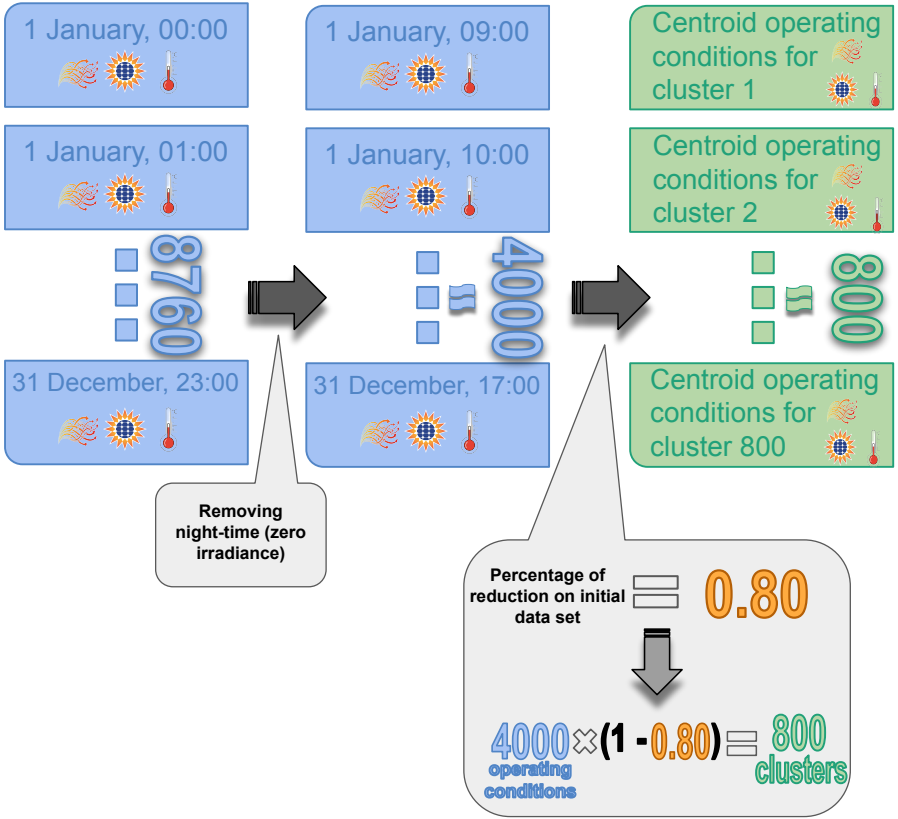
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Straight-Forward Aggregation (SFA)



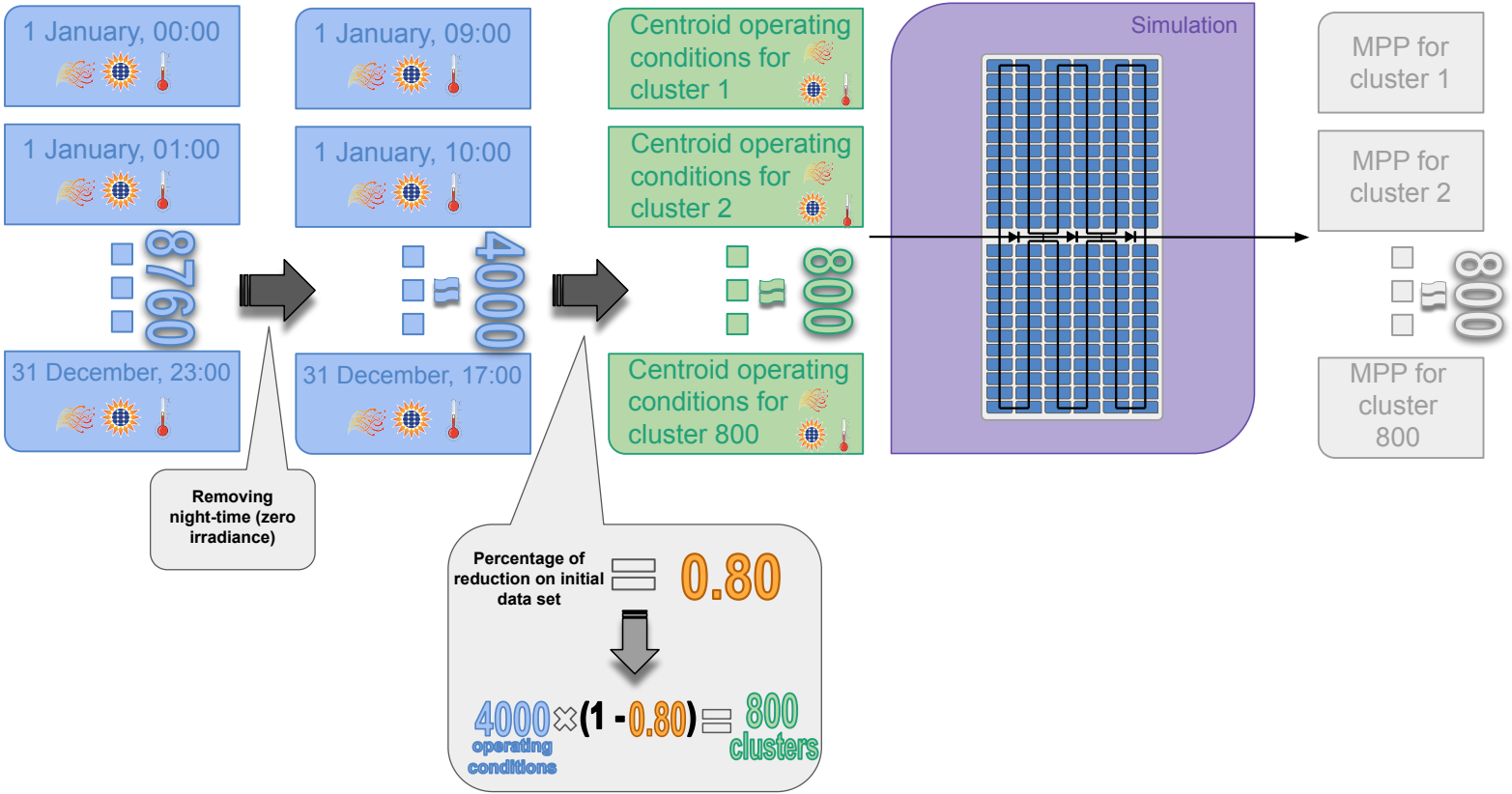
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Straight-Forward Aggregation (SFA)



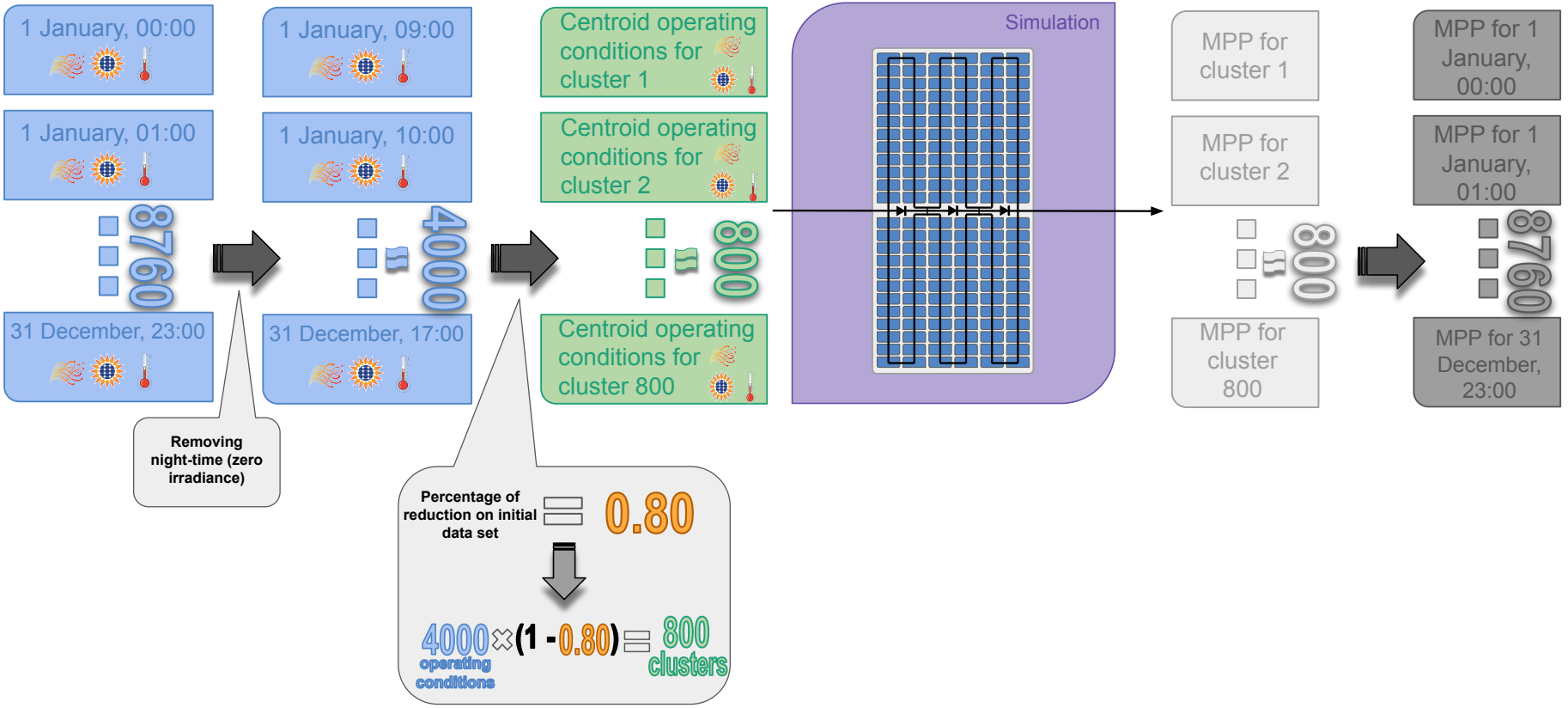
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Straight-Forward Aggregation (SFA)



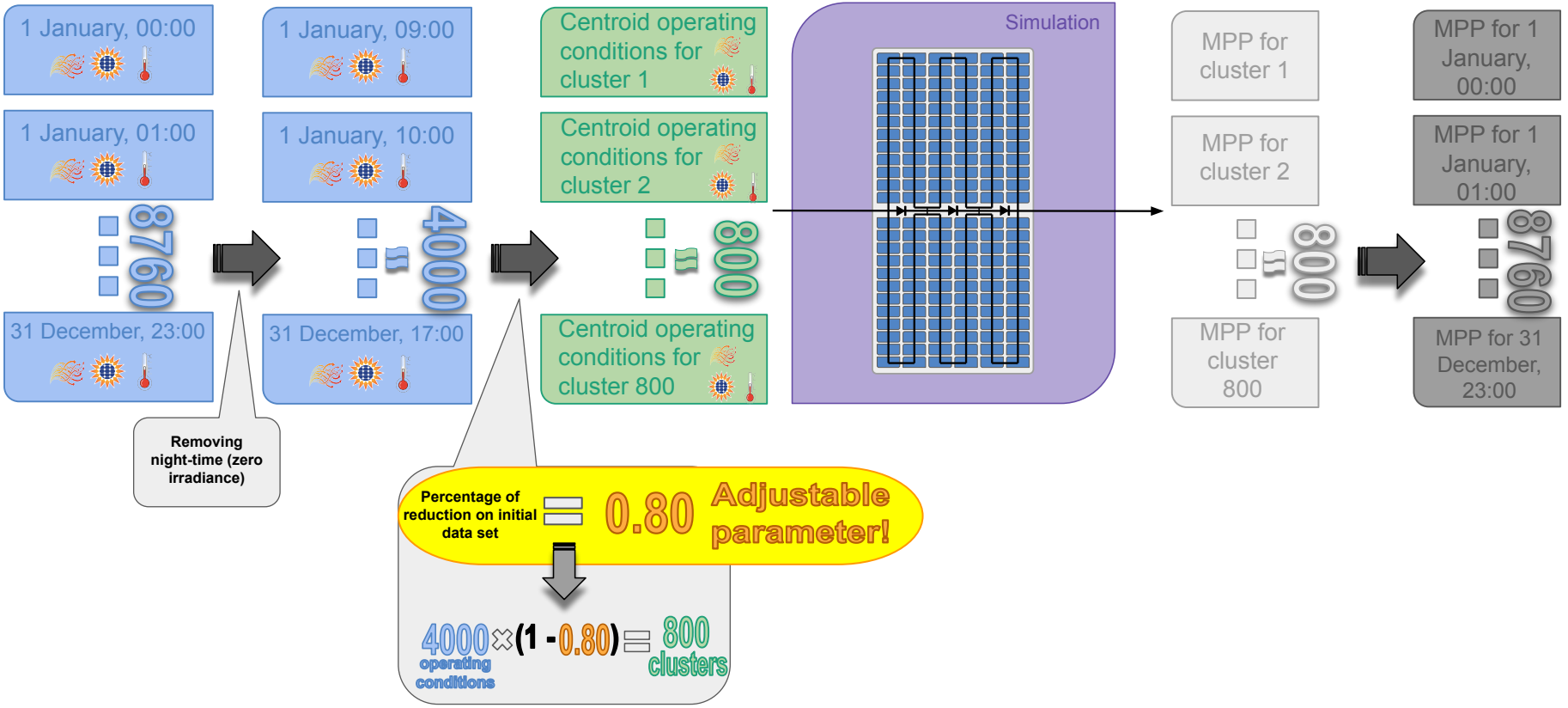
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Straight-Forward Aggregation (SFA)



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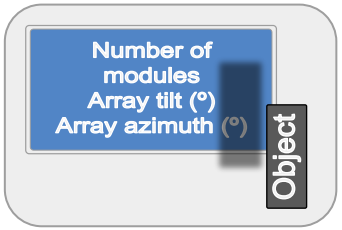
Straight-Forward Aggregation (SFA)



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Simulation scenarios for performance testing

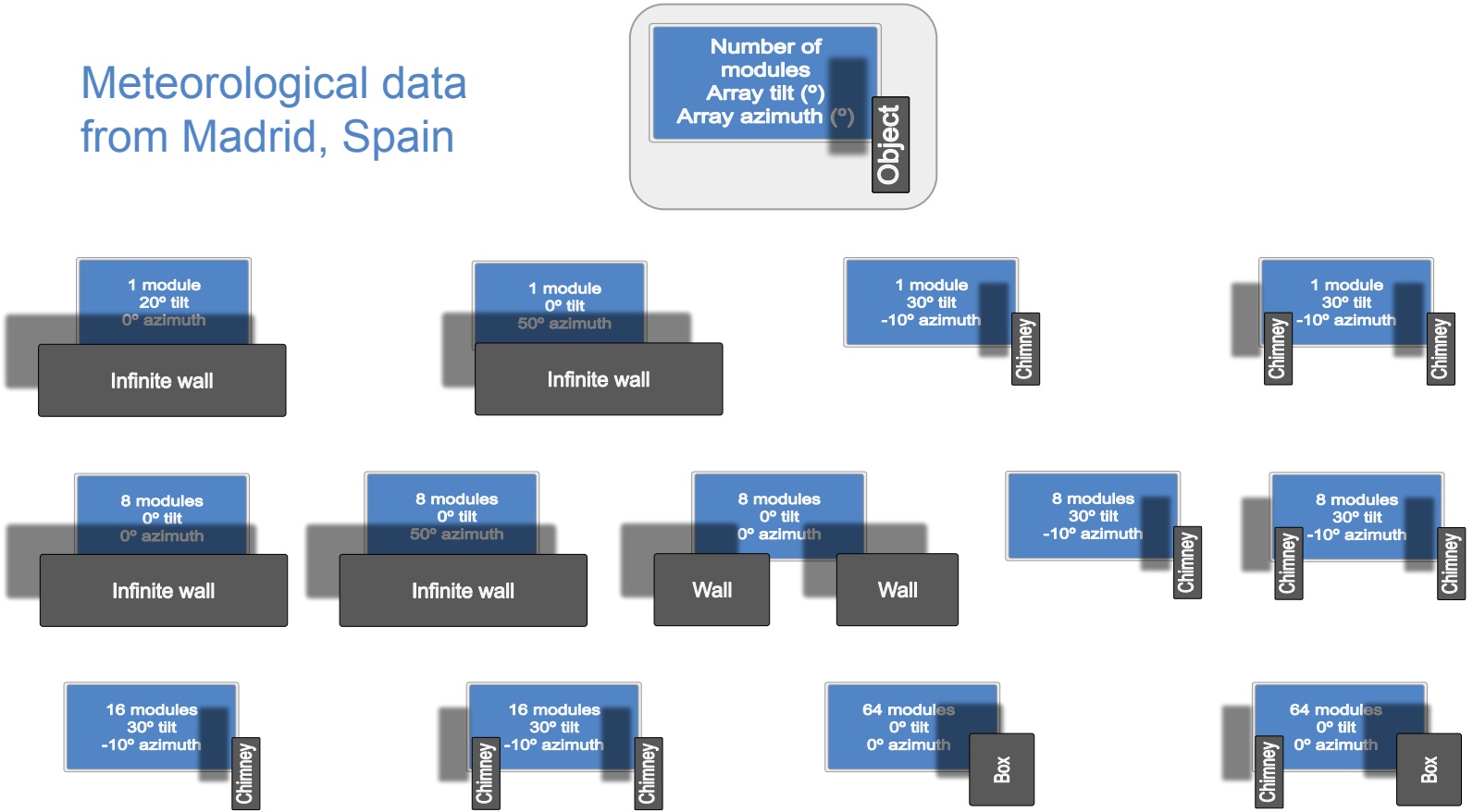
Meteorological data
from Madrid, Spain



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Simulation scenarios for performance testing

Meteorological data
from Madrid, Spain



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SFA performance results

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

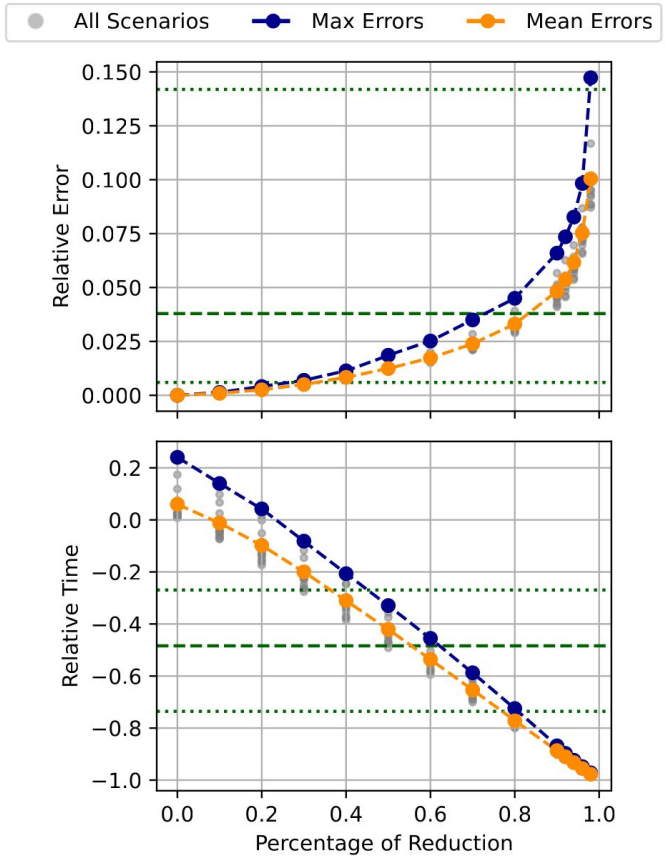
$$Relative\ Time = \frac{time_{new\ method} - time_{reference\ method}}{time_{reference\ method}}$$

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SFA performance results

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

$$Relative\ Time = \frac{time_{new\ method} - time_{reference\ method}}{time_{reference\ method}}$$

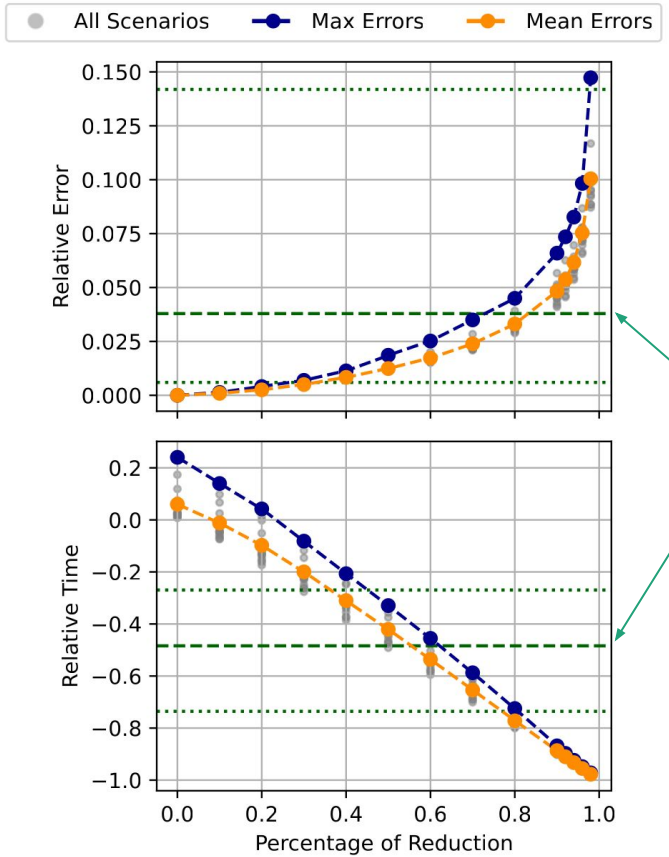


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SFA performance results

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

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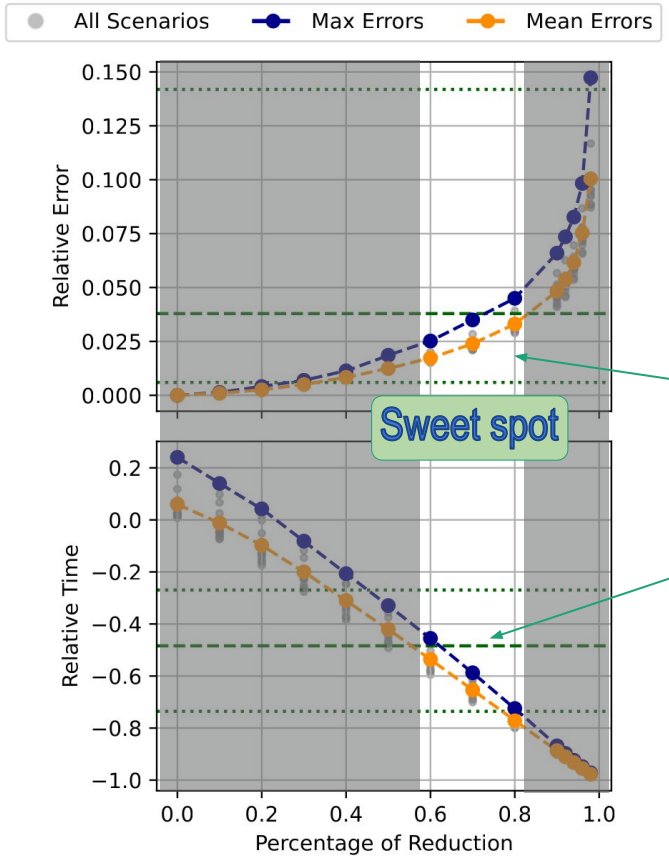
Module-level performance for comparison

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Less errors than module-level simulation

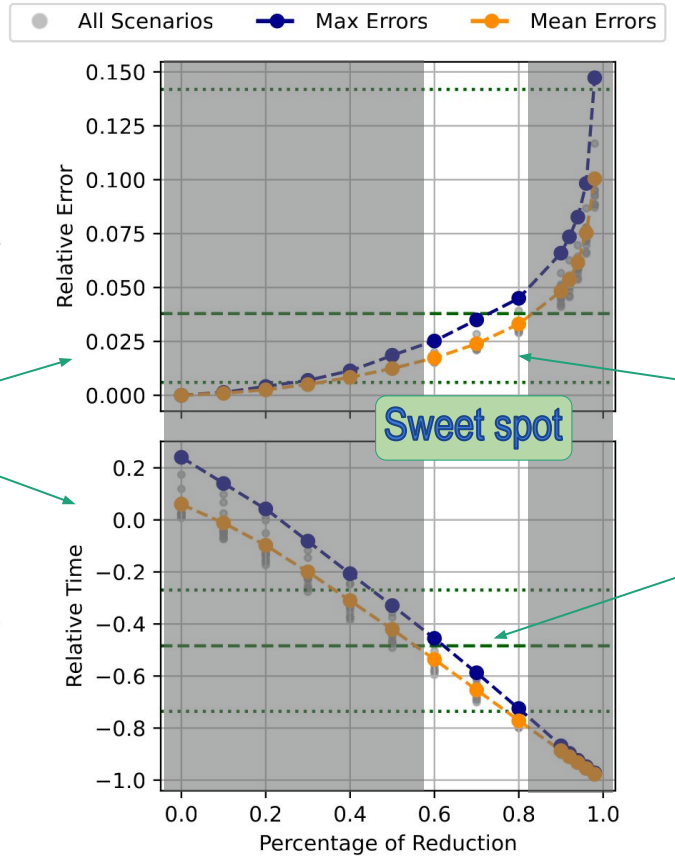
More reduction in computational time than module-level simulation

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SFA performance results

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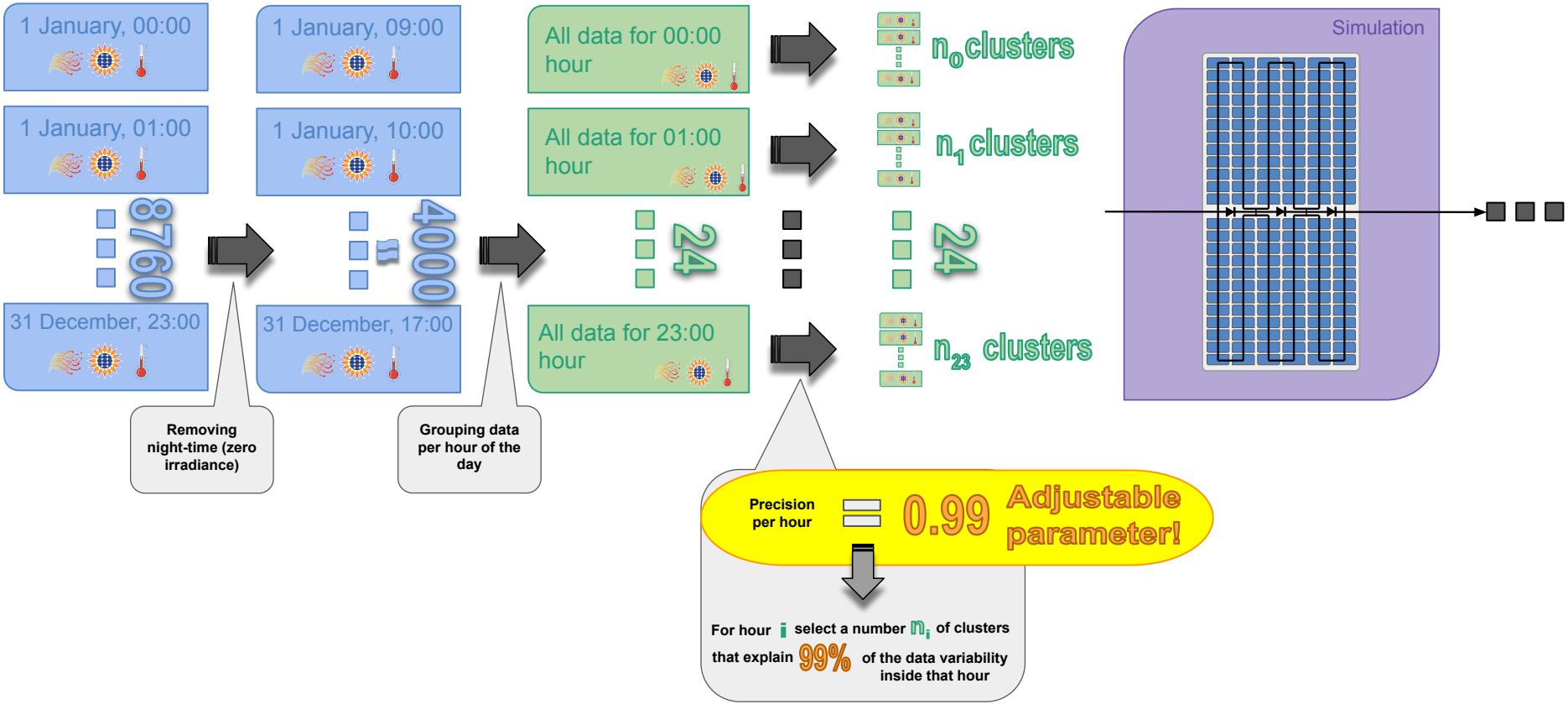
More stable results!

Less errors than module-level simulation

More reduction in computational time than module-level simulation

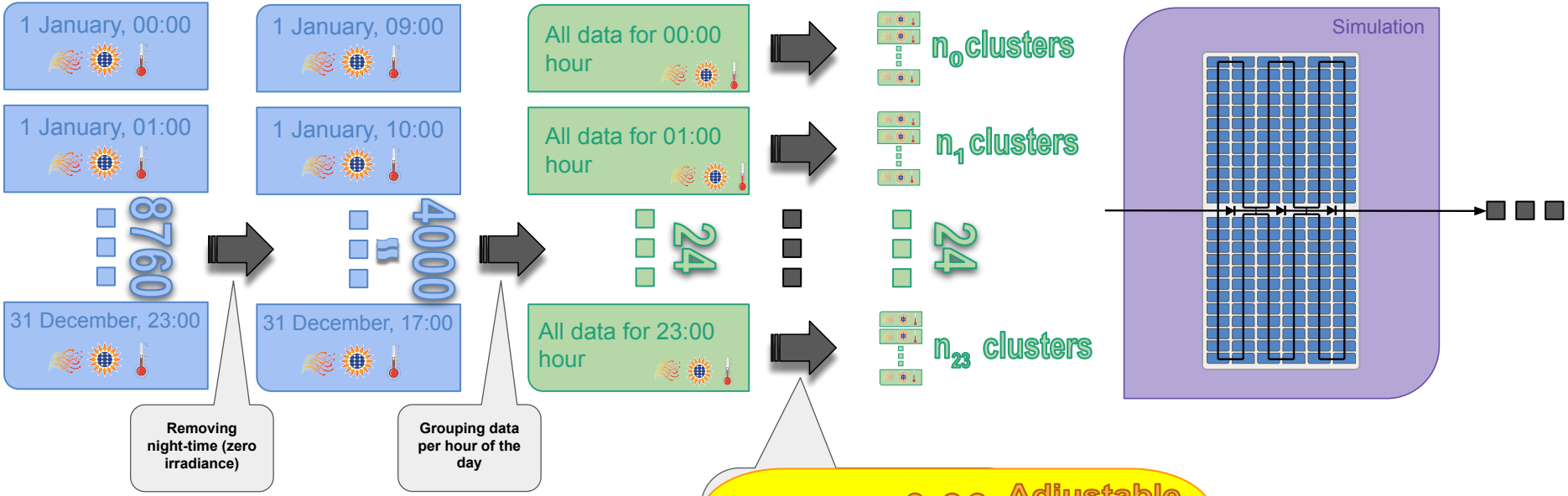
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Hierarchical Hourly Aggregation (HHA)



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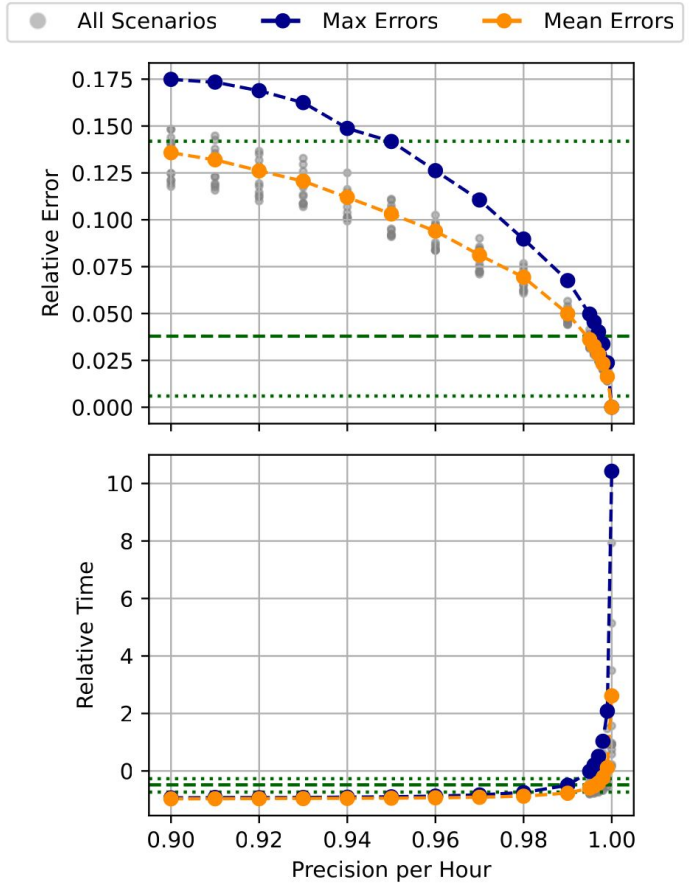
Hierarchical Hourly Aggregation (HHA)



more complex and less interpretable way of selecting the number of clusters compared with SFA

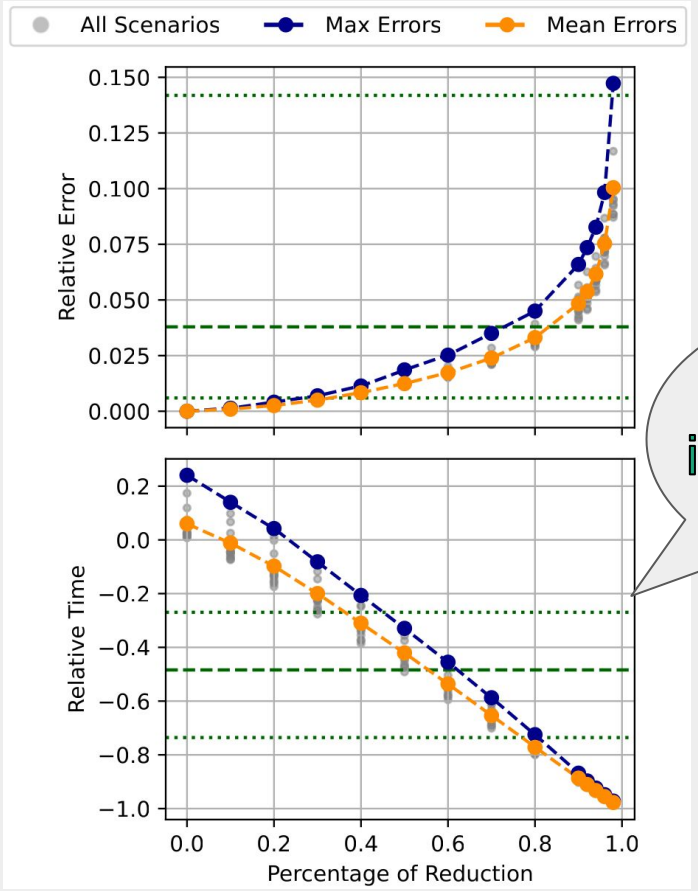
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HHA performance



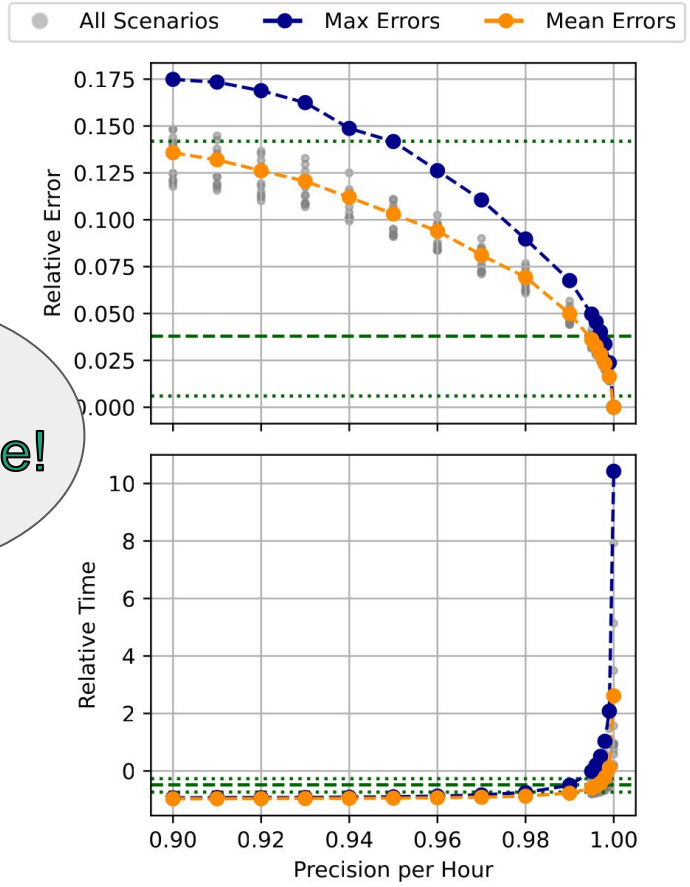
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SFA performance



More interpretable!

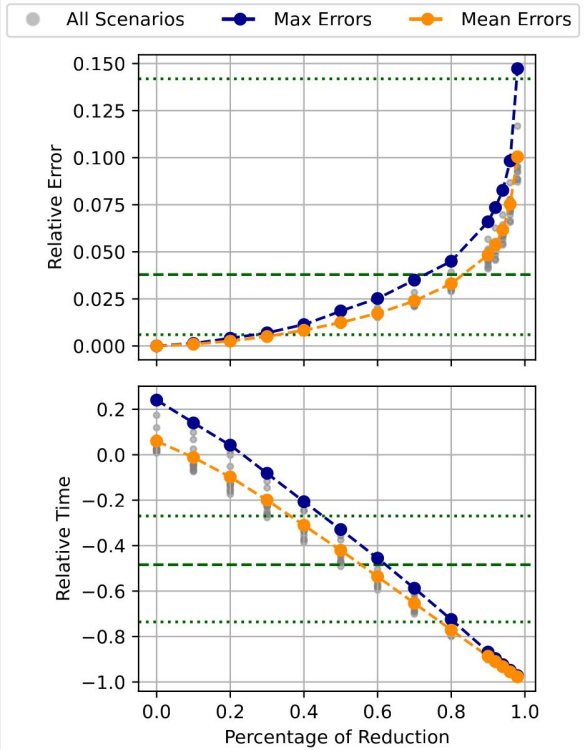
HHA performance



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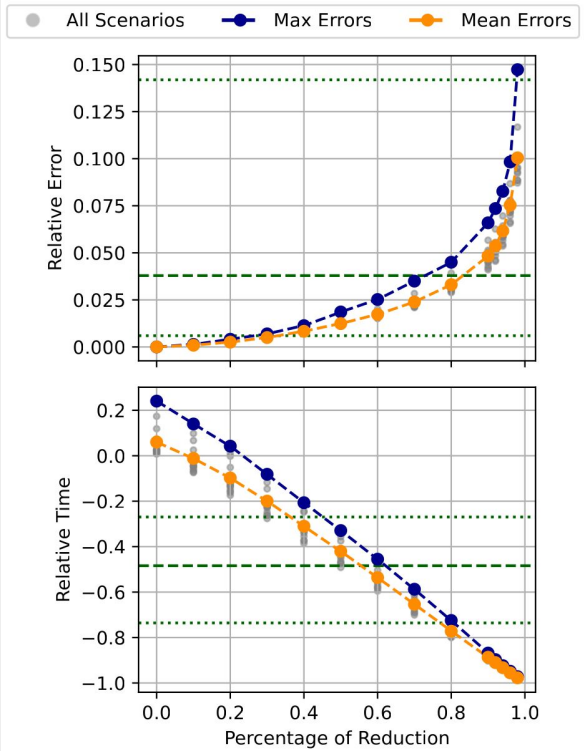
How do we compare both methods?

How do we compare both methods?



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How do we compare both methods?



mean

$$Relative\ Error = \frac{\sum \left| MPP_{new\ method} - MPP_{reference\ method} \right|}{\sum MPP_{reference\ method}}$$

VS

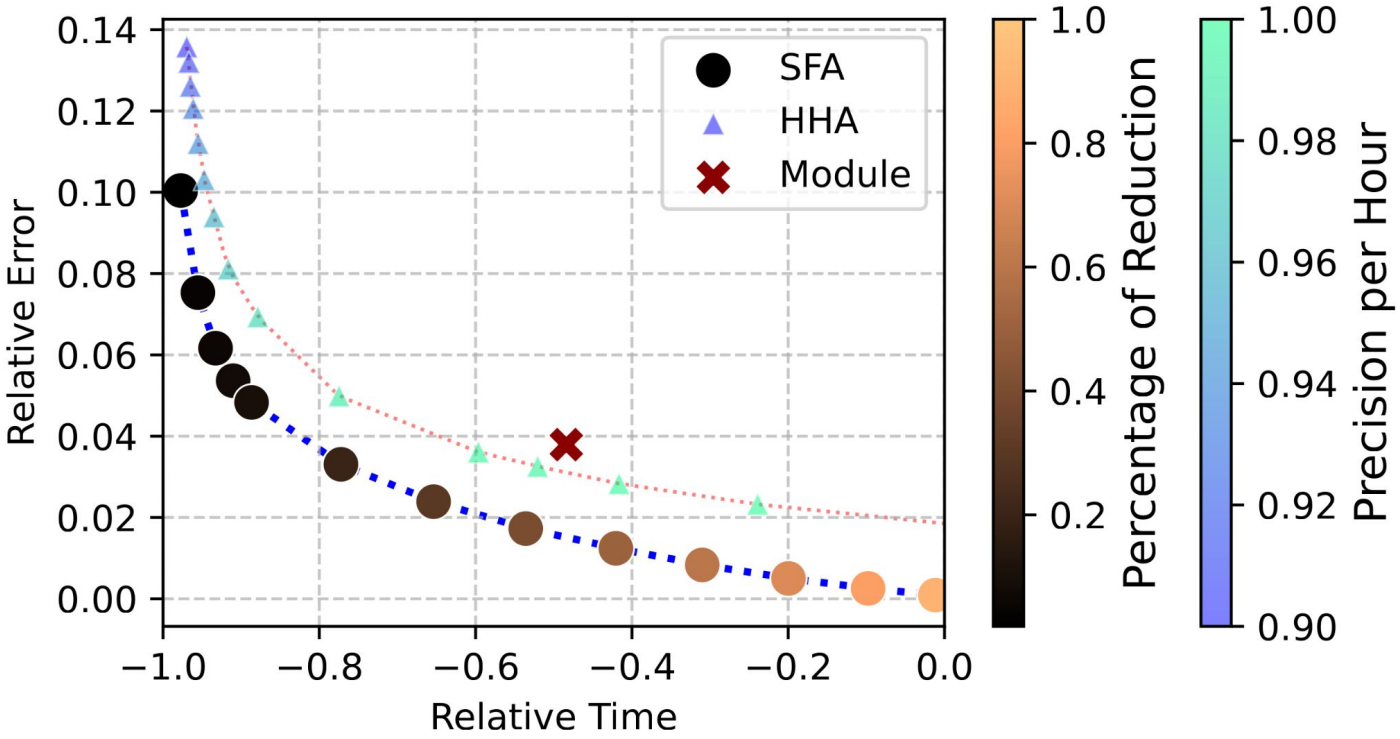
mean

$$Relative\ Time = \frac{time_{new\ method} - time_{reference\ method}}{time_{reference\ method}}$$

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Methods comparison (Pareto Frontier)

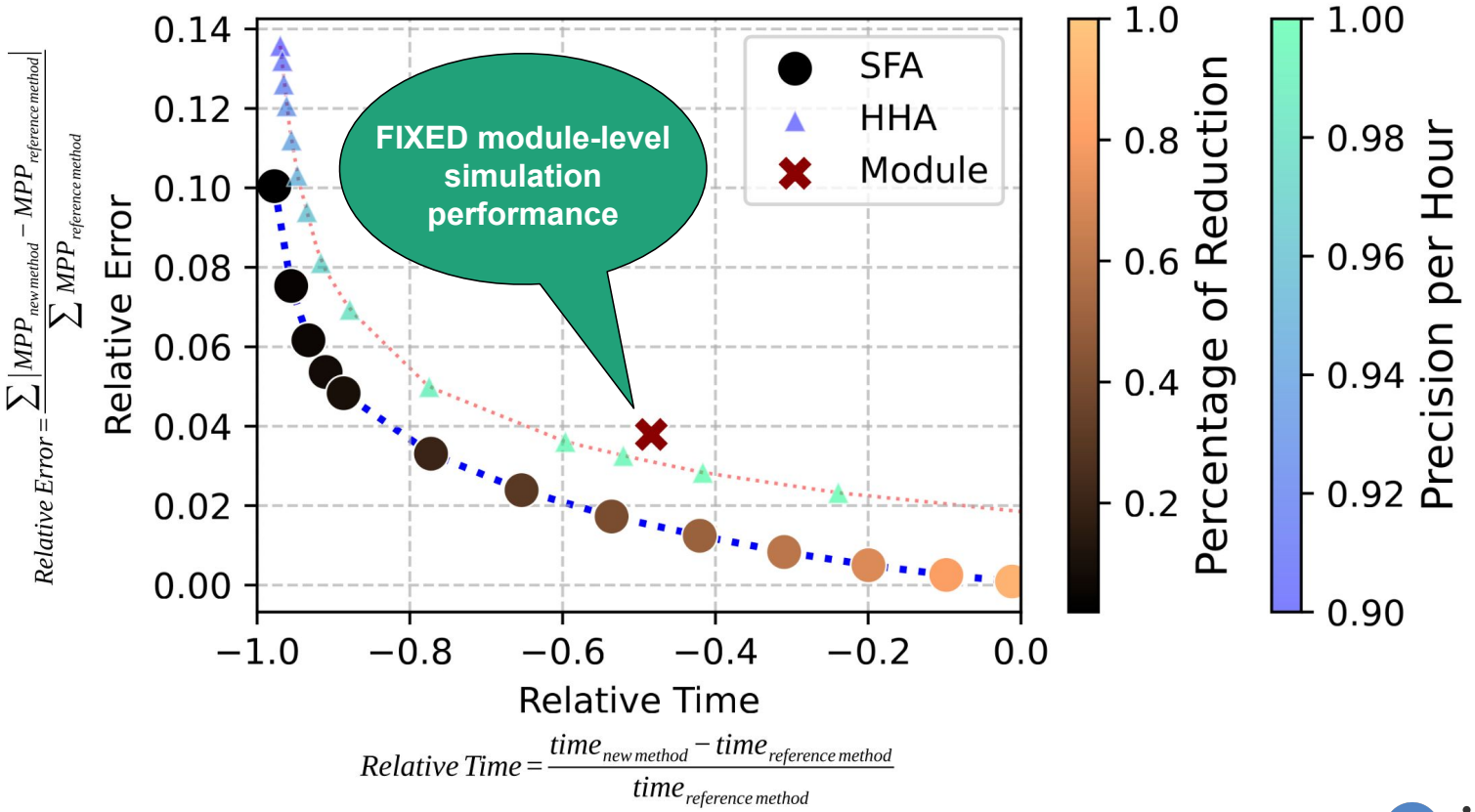
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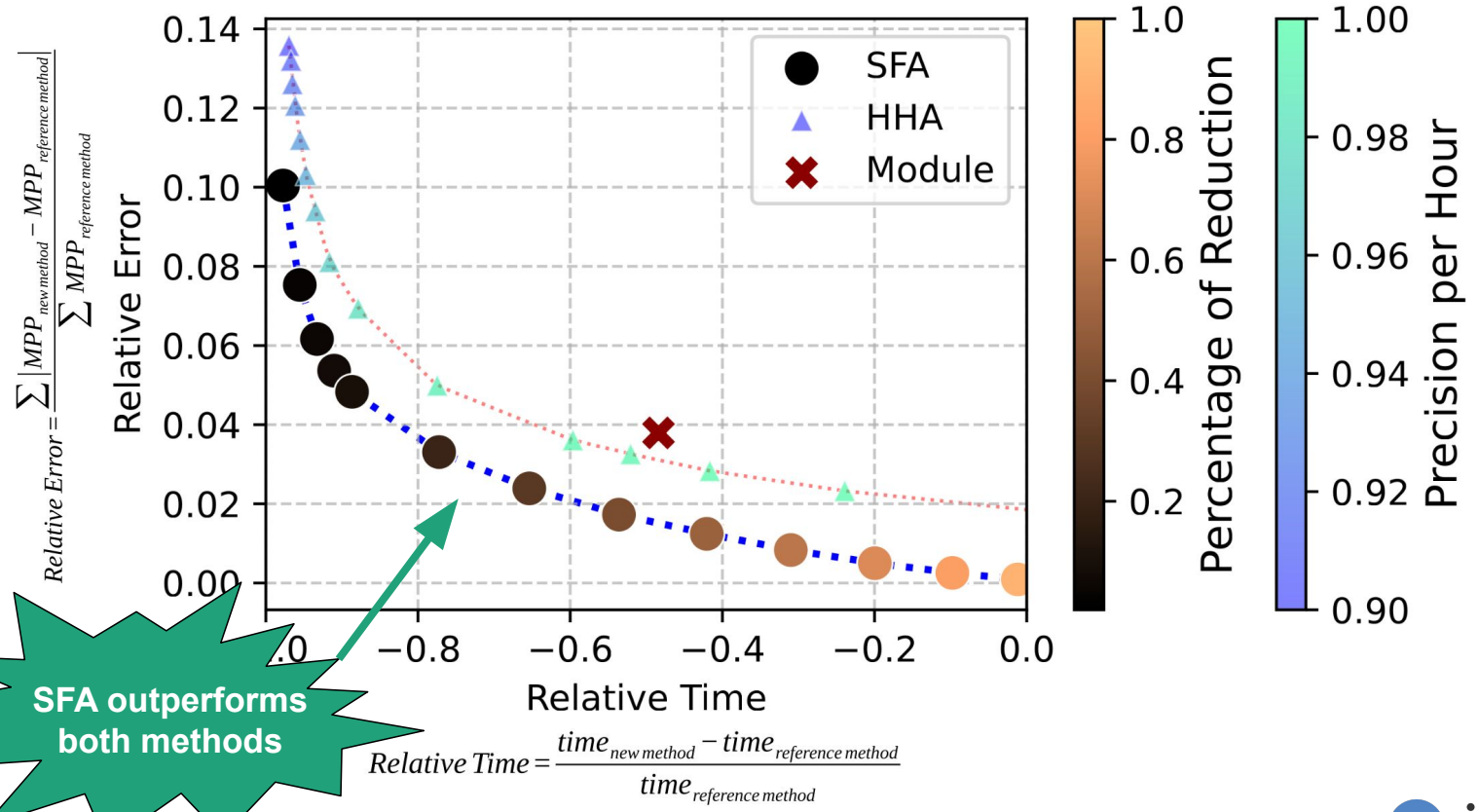
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Methods comparison (Pareto Frontier)



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SFA outperforms both methods

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Performance in terms of total annual power loss caused by partial shading (SFA)

$$MPP_{loss} = MPP_{no\,shading} - MPP_{shading}$$

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$$= \frac{\sum MPP_{loss\,new\,method}}{\sum MPP_{loss\,reference\,method}} - 1$$

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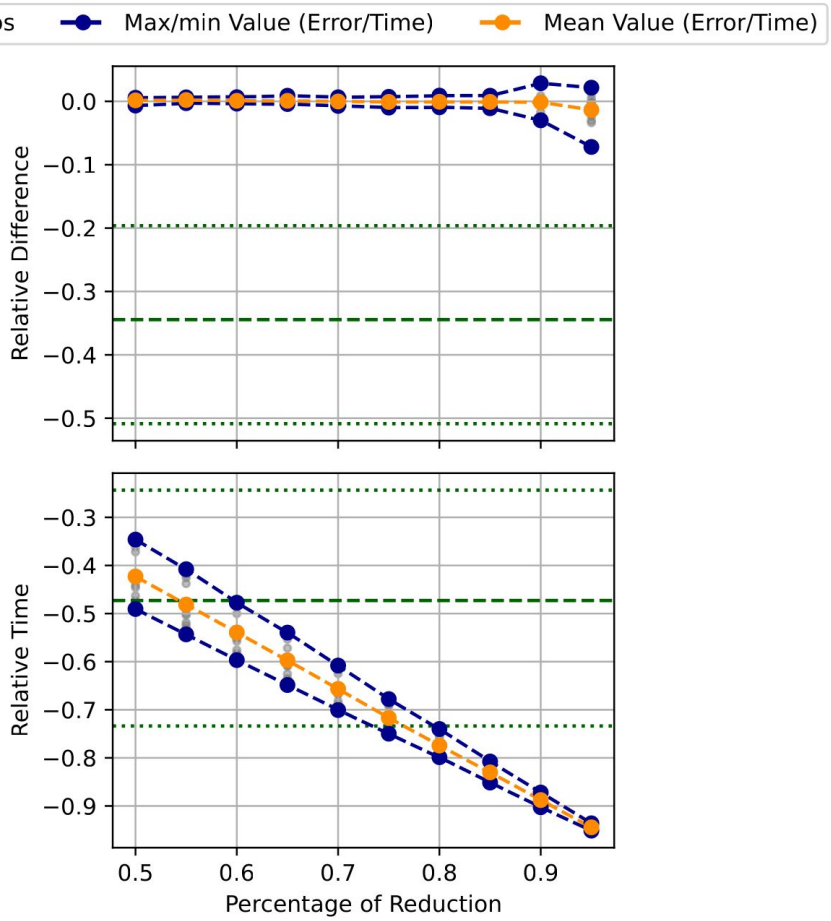
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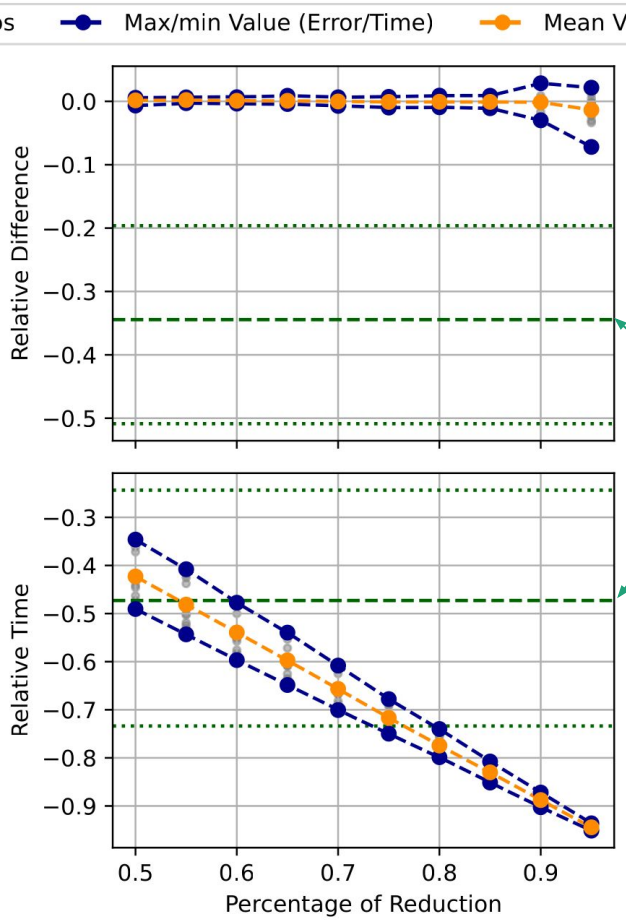
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Module-level performance for comparison

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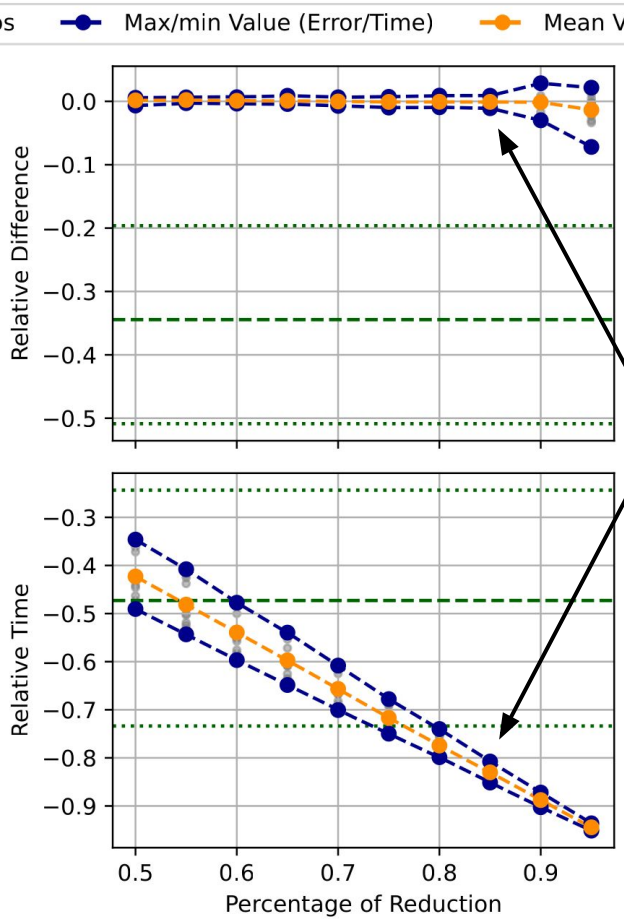
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$$Relative\ Time = \frac{time_{new\ method} - time_{reference\ method}}{time_{reference\ method}}$$



Simulation 85% faster with nearly the same results!

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Summary

- Aggregation methods for initial data segmentation

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- Better alternative in terms of accuracy/speed trade-off than module-level simulation without aggregation

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- Aggregation methods for initial data segmentation
- Better alternative in terms of accuracy/speed trade-off than module-level simulation without aggregation
- Very tunable models depending on the use case

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END

PV shading models with pvlib

Accelerating photovoltaic system simulations via statistical data aggregation